

The Free Lever and the Measured Walls: Re-Rounding, Whitened Low-Rank Correction, and the Limits of Post-Training Quantization Repair in llama.cpp

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Abstract

Consumer GPUs are compute-rich and memory-poor: an RTX 5090 pairs a datacenter-class processor with 32 GB of VRAM. We measure, on production llama.cpp artifacts, what post-training repair of aggressive quantization can and cannot buy under that constraint. The premise is a measured inversion: batch-1 decode is memory-bandwidth-bound, so a 27B model taken from 23.6 GB to 13.0 GB of VRAM decodes *faster* (61 to 99.7 tok/s), and accuracy is the only price of compression [rollup]. We contribute one lever that costs zero bytes and an audit of the levers that do not pay. The free lever is frozen-grid re-rounding: keep llama-quantize’s per-block grids byte-for-byte and re-choose only the codes under a captured activation-covariance metric. At identical bytes it improves a Q2_K model by 15% perplexity and 34% KL divergence at 0.8B (30 to 45 paired standard errors) and by 13% KL divergence at 27B ($t = -7.0$, $n = 100$), and emits a standard GGUF that runs on every deployed llama.cpp build [E24, E27]. Its caveat is measured: calibrated on one corpus it transfers to nearby unseen text but is worse than the untouched model on a distant one, and mixed calibration repairs both [wins]. The walls: weights and their quantization residuals are energy-full-rank; whitened low-rank correction is real at every scale, but its capture falls with width as a fitted power law rather than as r/d ; and at 27B, with every artifact one quantization step from BF16 and paired per-chunk statistics, the corrected configuration beats the adjacent ladder rung ($t = -2.85$) and loses to a pure-quantization mixed-type control built at the same bytes ($t = +2.62$). Three different low-rank mechanisms are indistinguishable at equal bytes; only type allocation escapes. On a mixture-of-experts model the correction loses a trivial promotion control at 6.9 paired standard errors. Against the released GSQ-RCO checkpoints of a 27B model, a 13-minute one-shot correction wins KL divergence at both rungs on two corpora, yet the trained artifact wins long-context perplexity and a 112-unit agentic coding suite against its own untrained baseline (net +12 units, $p = 0.012$): short-context divergence from the reference does not rank artifacts across refinement families [E32]. We also report a fused-kernel series that removes most of the adapter decode tax (+79% at 0.6B, +2.7% at 27B), an allocator fix worth +19.6% decode to any CUDA llama.cpp user, and the measurement protocol — paired statistics, interpolation controls, single-step provenance, pre-registration — that an adversarial review of our own first claims forced on us.

1. Introduction

1.1 The compute-rich, memory-poor regime

Current GPU economics gave consumers a strange machine: processors of datacenter caliber strapped to VRAM pools sized for last decade’s models. An RTX 5090 offers Blackwell tensor throughput behind 32 GB of memory; the models people want to serve on it are tens to hundreds of gigabytes at training precision. The classical framing of model compression for such hardware — trade compute time for memory by reconstructing compressed weights on the fly — turns out to be empirically backwards at the batch sizes consumer serving actually runs. Single-stream decode is memory-bandwidth-bound: every token reads the full weight set, and arithmetic units idle while HBM streams. We measured the consequence directly on a production 27B model (Qwen3.6-27B; the served checkpoint is “heretic-v2”, a community finetune of the base model with its multi-token-prediction layers preserved; the quantization and correction machinery is checkpoint-agnostic): the Q6_K reference occupies 23.6 GB of VRAM and decodes at 61 tok/s; the same model at Q2_K occupies 13.0 GB and decodes at 99.7 tok/s [rollup]. Smaller weights are not merely feasible — they are *faster*. The trade was never compute-for-memory; it is accuracy-for-memory, with compute as the free resource available to buy accuracy back.

That inversion defines this paper’s question: given that compute is free and bytes are everything, *how much quality can post-training machinery recover per byte — or per zero bytes — on top of a mature quantization stack?* We answer it on llama.cpp, the dominant consumer inference engine, whose K-quant and I-quant ladder (imatrix-weighted per-block quantizers spanning roughly 1.5-6.5 bits per weight) is the strongest deployed baseline in this regime and, as we will show repeatedly, a far harder rival than the literature’s round-to-nearest strawmen.

1.2 What this paper is not

This is not a paper claiming low-rank factorization compresses LLMs. It does not, and we can show exactly why and where. A full singular-value census of every tensor in our smallest subject finds that capturing 95% of Frobenius energy requires 50-80% of full rank on every projection type — compression at that rank saves approximately nothing, and for V/FFN projections it costs memory [E1]. The quantization *residual* $R = W - \text{dequant}(Q)$, the object the quantization-error-correction lineage (ZeroQuant-V2/LoRC, [1]; LQER, [2]; QERA, [3]; EoRA, [4]; CALDERA, [5]) corrects, is even less structured: near-isotropic, with 90% of its energy needing ~62% of full rank [E3]. Both nulls confirm the Marchenko-Pastur picture LQER published, and RILQ’s published diagnosis of the same wall — 2-bit quantization error is inherently high-rank, and layer-local low-rank correction saturates ([6]): unweighted SVD capture of a quantization residual is approximately r/d , i.e. nothing at deployable ranks.

Neither is this a paper claiming a new quantization method beats llama.cpp’s ladder. Our strongest corrected configuration at 27B, re-derived with every artifact one quantization step from BF16 and scored with paired per-chunk statistics on 100 chunks, beats the adjacent rung Q3_K_S (KLD $t = -2.85$, NLL $t = -5.75$, top-1 $t = +2.04$) and loses to a pure-quantization interpolation control built at the same byte point (KLD $t = +2.62$); the I-quant rung IQ3_XS scores better than both at fewer bytes, adapter-free. The history of that comparison is itself a result: the legacy unpaired 1σ “margin” flipped sign under clean provenance, read as a dead tie at 20 and 40 chunks, and resolved only at 100 [E27]. We state this plainly because our own first drafts did not: an adversarial review of the campaign’s claims [review] removed “first ladder victory” framings, two-point “laws,” and unqualified compression multiples, and the corrected framings — a rung beaten but not the interpolation line, wins-from-below, priced serving taxes — govern every sentence here.

1.3 What this paper contributes

(1) A zero-byte quality lever, conditioned on calibration breadth. llama-quantize picks good per-block grids but rounds every element independently. Freezing those grids byte-for-byte and re-choosing only the codes under a captured full activation-covariance metric — a closest-vector-problem sweep in the GPTQ/Babai lineage ([7]) — to our knowledge the first *one-shot*, *whitened* code re-optimization on llama.cpp’s deployed K-quant superblock hierarchy: GSQ ([8]) re-optimizes the same shipped format by stochastic search at GPU cost (the head-to-head is §4.3c), ReQuant ([9]) runs the same objective’s coordinate descent outside the GGUF format, and SchurQuant ([10]) refits scales jointly with codes out-of-format — placements proved exact in [theory-L6] — improves Qwen3.5-0.8B at Q2_K by -15% perplexity, -34% KL divergence, and +6.3 points top-1 agreement at *identical bytes*, resolved at $t = 30-45$ in paired per-chunk statistics against a BF16 truth reference [E24] at 27B the same sweep is worth -13.3% KL divergence and +1.07 points top-1 at identical bytes ($t = -7.0$, $n = 100$). The output is a byte-compatible standard GGUF: it runs unmodified on every llama.cpp build and every downstream application, with zero adoption friction. Two honest bounds. First, it recovers roughly half of the quality gap to the next quant tier, not the tier itself; gains concentrate at 2-3-bit tiers and shrink toward Q4. Second — measured, and load-bearing for any deployment claim: the gain is *given sufficiently broad calibration*. The wikitext-calibrated artifact improves an unseen prose corpus but is worse than bare on a held-out code corpus (PPL 3.470 vs 3.083), and a mixed 2:1 calibration improves both corpora at a modest on-distribution cost (§3.1) [wins]. Millions of deployed K-quant files carry this headroom today; within the re-optimization class named above, we are not aware of another tool that is simultaneously in-format, one-shot, and whitened (our tracker sweeps have missed tracker-resident prior art before, and the sweep record is disclosed as incomplete).

(2) A paired-statistics audit of every within-paradigm correction lever, at three dense scales and one MoE. Metric quality (diagonal versus full-covariance whitening), quantizer-corrector co-design (alternation, with a termination certificate), factor density (Q8_0 factors: capture-per-byte doubled free), rank allocation (measured sensitivity helps; energy water-filling is actively harmful), structural variants (column patches, shared bases, task

conditioning), and base-format composition (K-quants, I-quants, IQ1_S, NVFP4). Each lever is real; each shrinks with scale; and the audit’s negative results — the shared-basis null, the MoE control defeat, the anti-helpfulness of a misestimated Kronecker metric, the Lloyd fixed-point null and the equalization anti-result (§3.4, §4.6) — are its most statistically secure findings [review].

(3) Mechanisms for the failures. Fixed-rank whitened capture fell approximately fivefold when width grew fivefold across our first two diagonal-whitened scale points, in line with the rank-proportional-to-width bounds of CALDERA ([5]) and the intrinsic-LoRA lower bound ([11]). A regression over 1,146 tensors on four models replaces that two-point reading with a fitted form: $\text{capture} \approx a \cdot r^{0.68} \cdot d^{-0.76}$ under diagonal whitening and $a \cdot r^{0.37} \cdot d^{-0.52}$ under the full Gram, with the linear r/d rule the worst of six candidates and no width-invariant outlier head visible in the spectra (§4.2). We do not call it a law: the width axis is a single step [E33]. The design rule the campaign distilled — *correct where geometry favors correction (rank a large fraction of width); spend base bits where it does not* — is validated constructively by a geometry-aware mixed base at 27B and destructively by the IQ1_S asymptote experiment, where even 2.5 GB of full-covariance factors cannot substitute for carrier bits [E15].

(4) The systems layer that makes correction servable. llama.cpp’s stock LoRA path prices any adapter at a 20-40% decode tax that the bare quant rungs never pay. A three-phase fused-kernel campaign eliminates most of it (+44%, +62%, +79% adapter decode at 0.6B; +2.7% at 27B under an interleaved $N = 10$ protocol, against a measured +4.9% ceiling), yielding two transferable CUDA idioms — prefetch static operands before the grid dependency sync, and keep epilogues register-neutral by parking state in shared memory — and, en route, an allocator-level concurrency fix worth +19.6% decode to any CUDA llama.cpp user with graph optimization, adapter or not [E14, E25].

(5) A head-to-head with a trained competitor, and a warning about our own instrument. Against the released GSQ-RCO checkpoints of Qwen3.8-27B, a one-shot whitened correction computed in about 13 CPU-minutes wins KL divergence at both rungs on two corpora. The same experiment shows the limit of that instrument: the trained artifact scores below its own untrained baseline on 512-token KL divergence and above it on 2048-token perplexity and on an agentic coding suite (§4.3c).

(6) A measurement protocol earned the hard way. Calibration-distribution evaluation, missing interpolation controls, unpaired sub-2-sigma margins, and GPU boost drift each produced a headline our review subsequently retracted. The resulting protocol — single-step provenance from BF16, paired per-chunk statistics for every margin inside 2 sigma unpaired, ladder comparisons that include the mixed-quant interpolation control, serving tax priced beside every byte-fair quality claim, and timestamped pre-registration — is stated in §2.5 and applied to all results after the review [protocol].

1.4 Relation to prior work, in one paragraph

Our pipeline is deliberately built from published mathematics: the whitened closed forms are QERA’s ([3]) Theorem 1/2 pair, whose diagonal special case is literally llama.cpp’s imatrix statistic; the alternation protocol descends from LoftQ ([12]) with CALDERA’s best-iterate practice; the re-rounder is GPTQ’s sweep on a frozen grid, the setting where Bid-Up ([11]) proves monotonicity. What is new is where it is measured and what that measurement shows: same-pipeline full-versus-diagonal and correction-versus-ladder ablations on llama.cpp’s production K/I-quants at 0.6B, 0.8B, 27B, and a 35B MoE, against the strongest deployed baseline rather than round-to-nearest — including, uniquely in this lineage, *mixed-quant interpolation controls* and I-quant rivals at matched bytes, comparisons none of CALDERA/LQER/EoRA report against their RTN/GPTQ baselines (and which, at 27B and MoE scale, our own corrections lose); the frozen-grid re-rounder as a deployable tool class; the MoE expert-orthogonality negative; the fused-GEMV serving path the field’s own artifacts conspicuously lack (EoRA ships unfused in HF/vLLM; MiLo, [13], states its compensator is unfused; CALDERA’s Table 6 records the decode loss). The upstream conversation this feeds — llama.cpp discussion #8831, where an LQER-style correction was proposed in 2024 and predicted not to help mature quants (ik_llama.cpp #15) — receives here a systematic empirical answer: the skeptic was mostly right about correction versus the ladder, and wrong about the grid itself, whose codes carry a free 13-34% of KL divergence.

The paper follows the pipeline: method (§2), the free lever (§3), the walls (§4), the systems layer (§5), limitations and open work (§6), related work (§7).

2. Method

The pipeline has four stages: capture second-moment statistics on calibration text (§2.1); fit whitened corrections in closed form (§2.2) and/or re-round the base’s codes on their frozen grids (§2.3); optionally alternate quantizer and corrector (§2.4); serve through llama.cpp’s existing adapter path or our fused kernels (§5). All quality claims follow the measurement protocol of §2.5.

2.1 Statistics capture: activation Grams, MoE extension, gradient Grams

Input-side Grams. llama-imatrix already accumulates $\text{diag}(E[x^2])$ per weight column — the diagonal of the input second-moment matrix, and exactly the diagonal scaling LQER ([2]) and QERA-approx ([3]) employ. We extended it (a local, env-gated C++ patch; ~140 lines) to accumulate the full Gram $G = \sum x x^T$ per *distinct layer input*: `attn_k`’s input covers `q/k/v`, `ffn_up`’s covers `gate/up`, `attn_output` and `ffn_down` get their own — four Grams per layer. Accumulation is fp32 with a hand-rolled threaded rank-k update (no BLAS dependency, honoring llama.cpp’s no-third-party rule), written beside the normal imatrix. Validation: $\text{diag}(G)/\text{count}$ matches the stock imatrix to median relative error $2e-8$; exact symmetry; positive semidefinite [E10]. Inputs wider than 8192 dimensions are captured block-diagonally (8 blocks) — a memory-driven approximation whose consequences §2.2 makes explicit.

MoE extension. The `MUL_MAT_ID` (expert-routed) branch required its own accumulator: expert-slot-mixed rows must be unpacked per (token, slot) pair before the rank-k update, and `gate/up` stacks share a bitwise-identical routed input (verified), so one pooled Gram per stack input suffices. We capture *pooled* Grams — the union of routed inputs — rather than per-expert Grams, for a statistical reason as much as a storage one: at 16 calibration chunks, 65,536 routed pairs across 256 experts leave ~256 samples per expert for a 2048-dimensional Gram (rank-deficient by 8x), while the pooled Gram is full-rank; and the shared-basis hypothesis §4.5 tests is precisely a statement about the pooled distribution. Validation gates (symmetry exact, PSD, pooled diagonal matches the per-expert imatrix sums to $1.3e-4$, counts exact) all pass on Qwen3.6-35B-A3B [E23].

Output-gradient Grams. For two-sided (Kronecker-Fisher) metrics we built llama-gradmatrix, which runs llama.cpp’s own training loop forward+backward with learning disabled and captures $T = \sum g g^T$ per tensor, where g is the loss gradient at the pre-activation output row, hooked at the `OUT_PROD` nodes of the backward graph. Two structural repairs were required: llama.cpp’s trainer never backpropagates through the KV cache (`ggml_set_rows` has no backward, and the attention read is edge-disconnected from the write — independently found; upstream issue #21037 reports the same seam), which we bypass by feeding `k/v` directly to attention when the whole window is one micro-batch; and several ops on the hybrid delta-net architectures have no backward, handled by an explicit, disclosed gradient cut that stops flow through higher layers’ token mixing while keeping the residual path intact. All captures pass PSD/symmetry/count gates; the 0.8B capture’s gradient cut biases exactly the 18 linear-attention layers, a caveat that becomes load-bearing in §4.6 [E20].

2.2 Whitened low-rank correction: closed forms and the blocked-whitening caveat

Given residual $R = W_{\text{ref}} - \text{dequant}(Q)$ and an invertible whitener (L = damped Cholesky factor of G/count ; damping 0.01-mean-diag, the GPTQ/CLoQ convention, [7], [14]; the Cholesky-of-Gram whitening route is numerically viable at LLM scale by SVD-LLM’s theorem, [15]), the rank- r corrector minimizes the activation-weighted error $\phi(\Delta) = \|(R - \Delta)L\|_{F^2}$. The general two-sided weighted problem is NP-hard (Gillis-Glineur, [16]); the one-sided problem has the classical closed form: $\text{whiten } R_d = R L$, truncate, unwhiten. This objective is the $L_g = I$ special case of OBD-LLM’s two-sided whitened residual correction ([17]), and the one-sidedness is a defended choice rather than an omission: the tool path starts from a shipped GGUF where no gradients exist (backprop-free is the practical differentiator), the cheap diagonal end of output-side weighting is provably a no-op for the frozen-grid re-rounder (§2.3’s E24 theorem), and their measured two-sided gain is on weight decomposition at 8B, not residual correction on K-quant grids — the full defense and family placement are in [theory-L6]. We use the *projection form*: $B = U_r$ (top- r left singular vectors of R_d), $A = B^T R$. This is exactly optimal — for any invertible right whitener the unwhitening factor cancels, verified numerically to $7e-15$ against the explicit `argmin` [review] — and has the practical virtue that A is computed against the raw residual, so no ill-conditioned solve appears. Factors are computed by randomized SVD (Halko-Martinsson-Tropp, [18]; oversampling 32, 4 power iterations) with streaming per-tensor dequantization, making a 27B extraction a ~270-second job.

The blocked-whitening caveat, stated where the label is used. For inputs above 8192 dimensions (27B ffn_down: 25,600) the captured Gram is block-diagonal, and the projection form is exactly optimal *for the block-diagonal surrogate metric*, not for the true $E[xx^T]$. A synthetic cross-correlated repro measured a 25% excess of true-metric loss for the blocked optimum versus the full argmin [review]. Every “full covariance” 27B row in this paper therefore means: full covariance at ≤ 8192 -dim inputs, 8-block-diagonal for ffn_down. End-to-end measurements are unaffected (they score the artifact), but the mechanism claim “the metric was the bottleneck” is untested against a genuinely full metric at that width.

Two-sided variant. With both S and T captured, $B = L_T^T U_r$ and A solved per block against L_S — matching the closed-form argmin to $1.3e-15$ — with a factor-balancing step (equalize $\|B_i\|$, $\|A_i\|$; product provably unchanged, drift 9e-16) that exists because unbalanced un-whitening once overflowed F16 to 9,548 infs and a NaN serve [review, journal].

Factor density. Factors ship as Q8_0 (per-32 fp16 scale + int8), which halves factor bytes at measured quality parity (r64-Q8 31.26 vs F16’s 31.29 at 0.6B; r128-Q8 = r128-F16 at 53% of the bytes) — the Delta-CoMe/ZeroQuant-V2 observation ([19], [1]) confirmed in this pipeline; capture-per-byte doubles for free [journal]. Mixed per-tensor ranks fold the runtime’s alpha/rank scaling into A exactly as llama.cpp’s adapter loader expects (verified against llama-adapter.h semantics).

Serving. The corrected model is {quantized base GGUF} + {LoRA-form adapter GGUF}: $y = Q \cdot x + s \cdot B \cdot (A \cdot x)$ is llama.cpp’s stock build_lora_mm path, so the artifact serves on unmodified llama-server — at the stock adapter tax \$5 prices and then removes. Capture, however, is *not* stock: every full-covariance number in this paper depends on our imatrix Gram patch, and gradient Grams on llama-gradmatrix. “Zero code changes” was true only of the diagonal v0.

2.3 Re-rounding as a closest-vector problem on frozen grids

A K-quant tensor is a lattice-like product code: per 16-element sub-block, a scale/min pair (the *grid*) plus low-bit *codes*. llama-quantize chooses both, rounding each element independently. We freeze the grid bytes exactly as shipped and re-choose only the codes to minimize the whitened objective — a constrained closest-vector problem on the frozen product grid. Our solver is the GPTQ error-feedback sweep, which is precisely Babai’s nearest-plane algorithm under the Cholesky factorization of the inverse Gram: process columns in order, round the current column on its frozen grid, and propagate the scaled error to unprocessed columns via the upper-triangular factor U ($\text{inv}(H) = U^T U$). Frozen grids place us in the setting where Bid-Up ([11]) proves per-sweep non-increase of the objective. ReQuant’s ([9]) frozen-grid coordinate refinement is exact cyclic coordinate descent on this same quadratic — its $-dq \cdot g + dq^2 \cdot \tilde{H}_{jj}$ score, with the gradient maintained by rank-one refresh, is the exact per-coordinate change of our objective at the current iterate (verified from their Eqs. 5-10) [theory-L6] — so the two tools differ in sweep schedule and format scope, not objective; SchurQuant ([10]) works the same full-Gram quadratic with the scale block unfrozen, alternating suffix-eliminating Schur-conditioned code descent (the chunk-block generalization of this sweep’s column-wise conditioning) with closed-form continuous scale/zero-point refits [same source, Claim 3].

Implementation facts that carried published numbers: the Q2_K codec was mirrored bit-exactly from gguf-py (byte-identical repack round-trip on real tensors); the triangular convention is load-bearing (a lower factor silently no-ops the feedback and reproduces llama-quantize’s codes bit-for-bit — diagnosed when PPL matched to four decimals); PD failures escalate damping rather than falling back to a dense factorization that would break sweep causality; lazy-batch blocking gives 8x speed at bit-identical output [E13]. The Kronecker (two-sided) generalization — a nested row-within-column sweep under $U_S \otimes U_T$, validated code-bit-exact against brute-force GPTQ on the explicit Kronecker factor — is derived in E24 together with a small negative theorem worth keeping: a *diagonal* output-side weighting is provably a no-op for frozen-grid per-element rounding (each row’s argmin is scale-invariant), so off-diagonal feedback is the only channel through which an output metric can move any code [E24].

The output is byte-compatible with the input format: same size, same type tags, servable by every existing binary. This is the property that makes §3’s result a *deployment* result and not just a benchmark.

2.4 Alternation and its convergence certificate

Correction quality depends on the base; the base’s grid choices depend on what the corrector will absorb. The alternation $Q_{t+1} = \text{quant}(W - C_t)$, $C_{t+1} = \text{corr}(W - Q_{t+1})$ — LoftQ’s template ([12]) with llama-quantize as a *black-box* quantizer and the whitened closed form as the exact corrector step — is run with best-iterate tracking and a stop-on-worse rule. The theory note [theory-alt] upgrades that rule to a certificate: (Claim 1) the quantizer’s reachable output set is finite, the corrector step is an exact argmin, so the best-iterate value is non-increasing over a finite set and becomes constant after finitely many rounds — termination in the exact sense that no further round can improve the tracked iterate; (Claim 2) round 1 improves iff pre-subtracting the corrector steers the quantizer to a grid point whose residual is more rank- r -compressible in the whitened metric — measured true at both scales (whitened capture $0.37 \rightarrow 0.46 \rightarrow 0.51$ across rounds at 0.6B). Claim 3, informal, explains the geometric decay of per-round gains via the composed map approaching a fixed point where no code crosses a decision boundary of the quantizer’s internal metric ψ — with one measured amendment: when the alternation was pushed to an 8-iteration cap at 0.8B (E26), *no* codes fixed point was reached within budget; 92/96 tensors entered a $\sim 0.1\%$ limit cycle at the floor, so what actually happens is best-iterate convergence onto a limit cycle, not a fixed point [E26]. The certificate’s best-iterate framing survives unchanged. The mismatch $\psi \neq \phi$ bounds achievable quality and motivates ProjQ-style metric surgery ([20]), with the Bregman-damped scheme of [21] as the proven-descent variant. In practice $T \leq 2\text{-}3$ rounds; deltas after round 2 are sub-noise.

2.5 Measurement protocol

Every result in §3-§5 either follows these standards or is labeled legacy [protocol]. (1) *Reference and provenance*: the reference is the specifically *targeted* artifact — BF16 wherever it exists, and where it does not (the MoE of §4.5, for which we had no BF16) the strongest vendor artifact, stated per table; every base and adapter derives from the reference in one quantization step; mixed-provenance rows never share a table without a provenance column. (2) *Metrics*: mean KL divergence and top-1 agreement versus reference logits (≥ 40 chunks; the 27B endgame of §4.3 runs 100; the MoE table and a few labeled legacy rows run 20, where they settle orderings by paired statistics only) with printed standard errors — KLD because it tracks answer flips at Spearman 0.981 ([22]) and is the llama.cpp maintainers’ preferred currency (discussion #4110) — with full-corpus perplexity secondary and a stated silent-zone caveat near baseline ([23]). (3) *Paired statistics*: any comparison inside 2 sigma unpaired is decided by the paired per-chunk difference (mean $\pm$ sem of Δ), recovered from llama-perplexity’s per-chunk streams; rival configs share eval tokens and reference logits, so pairing has materially more power than the unpaired sigmas our first-night claims leaned on. (4) *Comparisons*: byte-fair means total bytes (base+adapter) *and* stated VRAM; the ladder gets its full lineup — K-quants, I-quants, and the mixed-quant *interpolation control* at the matched byte point — at every scale. (5) *Speed*: same-session paired A/B only; boost drift documented (up to 9% between sessions on identical configs); stock versus fused-kernel numbers labeled; the serving tax always priced beside byte-fair quality claims. (6) *Pre-registration*: from the second review onward, every experiment’s configuration, pairs and decision rules were written to the journal with a timestamp before the run, and deviations are recorded. (7) *Claims*: “law” requires ≥ 3 scale points with a stated functional form; otherwise “observation.” Novelty claims require a tracker sweep, not only arXiv — two of our campaign’s “discoveries” (the MTP-imatrix blind spot; the KV-backward seam) were already reported upstream (#23476/#23575; #21037), and are handled here as independent confirmations.

3. Results I: the free lever

3.1 What re-rounding buys at zero bytes

The re-rounder changes nothing about file size, VRAM, or serving speed: it re-chooses which already-available code each weight gets, using one calibration pass. We report it at three scales, in order of scale.

Qwen3-0.6B (Q2_K-imat, legacy Q8_0-referenced scoring). The first measurement, and the largest effect [E13]:

config (296 MB, identical bytes)	PPL	KLD	top-1
Q2_K-imat bare (llama-quantize)	43.33	0.766	58.2%
1. re-round (free)	35.54	0.556	64.0%

An 18% perplexity improvement at identical bytes on an unmodified serving stack, resolved at ~ 20 sigma on PPL and ~ 19 sigma on KLD even unpaired [review].

Qwen3.5-0.8B (Q2_K, BF16-truth scoring, paired per-chunk statistics — the protocol-v2 flagship measurement). A second, hybrid-attention architecture, scored against the true BF16 reference with the paired-statistics repair applied throughout [E24]:

config (436 MB, identical bytes)	PPL	KLD	top-1
Q2_K bare	33.09	0.4902	66.4%
1. re-round, input-only (free)	28.09	0.3234	72.7%

Paired per-chunk deltas versus bare: mean $\Delta\text{NLL} -0.1637 \pm 0.0036$ nats/token ($t = -44.9$; better on 557/580 chunks), $\Delta\text{KLD} -0.1668 \pm 0.0055$ ($t = -30.3$; better on 40/40 chunks), $\Delta\text{top-1} +6.26 \pm 0.45$ points ($t = +14.1$). This is -15% PPL and -34% KLD for zero bytes, and it establishes that the 27B fade below was not a scale law (finding 1 of E24): the free lever transfers across architectures at full strength where its codec coverage is complete.

Qwen3.6-27B (Q2_K-imat; paired, $n = 100$; legacy-provenance pair). At production scale the lever is smaller and still resolved. On 100 chunks against BF16 truth logits [E27]:

config (10.86 GB, identical bytes)	PPL	KLD	top-1
Q2_K-imat bare	7.735 ± 0.126	0.1603 ± 0.0032	83.48%
1. re-round (free)	7.696 ± 0.124	0.1390 ± 0.0029	84.55%

Paired per-chunk: $\Delta\text{KLD} -0.0213 \pm 0.0030$ ($t = -7.02$; better on 77/100 chunks), $\Delta\text{top-1} +1.07$ points ($t = +4.72$), $\Delta\text{NLL} -0.005 \pm 0.005$ ($t = -0.98$, a tie). That is -13.3% KL divergence and one point of top-1 agreement for zero bytes; perplexity does not move. Two labels travel with the row. The pair is a requantization of a Q6_K file rather than a single step from BF16, so it is not tabled beside the single-step rows of §4.3 — both sides share that provenance, so the difference is clean. And the sweep covers only part of the model: inside a nominal “Q2_K” 27B file, llama-quantize promotes `ffn_down` and `attn_output` to Q3_K and `attn_qkv/attn_v` to Q4_K, and our sweep implements only the Q2_K codec. The unswept mass is most of the model’s sensitive tensors. The first reading of this row, at 20 chunks and unpaired, was inside its error bars (0.78σ); the 40-chunk paired reading gave $t = -3.70$ and the 100-chunk reading is the one reported. Q3_K/Q4_K codecs are open work (§6.3); returns should be smaller at higher bit-widths.

In summary: at small scale the lever recovers roughly half of the quality gap to the next tier at zero bytes (0.8B: 51.6% of the KLD gap, 51% of the PPL gap; 0.6B: 45% of the PPL gap); at 27B, with partial codec coverage, it recovers 13% of the base’s KL divergence. It is not “Q2 at Q4 performance,” and it is not demonstrated at tiers whose codecs we have not written [wins]. The consumer reading: quality bought at zero bytes is *memory* — a user who needed the next tier’s quality can sometimes stay a tier lower, which on 8-16 GB cards is a model class fitting or not.

The held-out arc: transfer depends on distance from the calibration corpus. The first held-out pass evaluated the wikitext-calibrated 0.8B re-round on a disjoint code corpus and found it *worse than bare*: PPL 3.470 versus 3.083 (bare), while the fc correction generalizes (2.649). Mechanism: re-rounding is a pure fit to the calibration covariance with no anchor beyond the grid — there is no bias term, no structure prior, nothing pulling toward the untouched solution off-distribution. A mixed 2:1 wikitext:code calibration (true-interleaved; a first sequentially-concatenated attempt never reached the code text and is documented as a bug) rescues both corpora: wiki 33.09 $\rightarrow$ 29.84 and code 3.083 $\rightarrow$ 2.785, at a modest on-distribution cost versus narrow calibration (-10% versus -15% wiki PPL against the 40-chunk baseline). Two pre-registered follow-ups complete the map [wins, E22]. On a third corpus in neither calibration set (technical prose), both re-rounded variants *improve* on the untouched model: bare 57.56, wikitext-calibrated 54.40 (-5.5%), mixed-calibrated 55.32 (-3.9%). And the mixed variant, scored on full metrics, improves both calibration-covered corpora against bare (wikitext KLD 0.490 $\rightarrow$ 0.372; code KLD 0.318 $\rightarrow$ 0.219 with +2.5 points top-1), giving up some on-distribution fit (the wikitext-only artifact reaches 0.323). The reading we hold: re-rounding transfers within a neighborhood of its calibration distribution and can damage a distant one unless that one is covered. Three caveats remain. The mixed-versus-narrow comparison changed corpus composition, calibration budget (+20%) and chunk sampling together; an accidental control bounds calibration-sampling variance alone at 0.55 PPL ($\sim 2\%$), which is larger than several 0.8B margins in this paper;

and the third corpus was assembled from our own literature notes, which are machine-drafted prose, so it is one sample of “unseen text,” not a benchmark. The full mix-ratio grid and the 27B held-out test of the re-rounder have not been run. Every deployment sentence in this paper therefore conditions the free lever on calibration breadth and a held-out gate.

3.2 The estimator result: a better metric on paper made it worse (E24)

The natural upgrade to input-only re-rounding is the Kronecker global metric $\text{tr}(T dW S dW^T)$ with both measured factors — YAQA’s Kronecker-factored Hessian ([24]) instantiated with our captured S (input Grams) and T (output-gradient Grams). The exact nested LDLQ sweep of §2.3 provably reduces that surrogate objective (verified: ~35% surrogate reduction on correlated synthetics, codes bit-exact against brute-force Kronecker GPTQ), and setting $T = I$ reproduces the input-only sweep bit-for-bit. The end-to-end result [E24]:

comparison (0.8B, paired)	ΔNLL	ΔKLD	$\Delta\text{top-1}$
kron vs bare	-0.1154 ± 0.0042 ($t = -27.7$)	-0.1193 ± 0.0081	$+5.58 \pm 0.50$
kron vs input-only	$+0.0483 \pm 0.0031$ ($t = +15.5$)	$+0.0475 \pm 0.0058$ ($t = +8.2$)	-0.69 ± 0.42 (unresolved)

The richer metric with our measured T is *anti-helpful*: it gives back a third of the input-only win, at 15 paired sigma. The solver is exonerated (it does exactly what the math says); the failure is the estimator. Our T comes from a single 16,384-token pass — the cheapest possible instantiation of a K-FAC factor — carries the delta-net gradient cut (§2.1) on exactly this architecture, and its late-layer K/V eigenstructure sits at the fp32 noise floor by the capture tool’s own gates. With a misestimated T , per-row Cholesky feedback actively relocates error into the wrong rows. The general lesson, which recurs at every layer of this paper (§4.6): *an approximation of the metric is not a monotone approximation of the outcome* — estimator quality gates metric quality, and one-sided or under-sampled proxies of a two-sided geometry are traps. YAQA proper spends far larger sketch budgets on exactly this estimation problem; our result measures what happens when one does not.

A pre-registered discriminating run asked whether the harm came from the gradient cut or from the estimator class. On Qwen3-0.6B, a classic architecture with no gradient cut and 196 exact gradient Grams, the same Kronecker sweep is worse than the input-only sweep *and worse than the untouched model*: PPL 50.53 / KLD 0.866 / top-1 58.0%, against 35.54 / 0.556 / 64.0% input-only and 43.33 / 0.766 / 58.2% bare. A cleaner capture made the harm larger. The model changed along with the capture, so the magnitudes are not a pure comparison, but the direction rules out capture coverage as the explanation: a single-pass Kronecker factor of this class hurts [journal].

3.3 The ordering rule: re-round for serving, alternate for correction

Where does the free lever sit in the co-design stack? Below the corrector, nowhere. Third paired-resolved confirmation at 0.8B: the re-round that is hugely positive bare (-0.164 nats paired) turns *negative* under a freshly-extracted fc corrector — bare+fc beats RR+fc at $t = +17.6$ (RR stack better on only 133/580 chunks), reproducing the 0.6B ordering (RR-then-correct 26.80 vs alternation-then-correct 25.27 PPL) and the ProjQ-deflated middle point (25.94) [E24, E13]. Mechanism: re-rounding minimizes *standalone* whitened error, spending grid resolution on directions the corrector would own, and leaves a less-correctable residual — measured as higher whitened capture of a worse-shaped residual (capture 0.78, worse KLD, on the 0.6B triad). Deployment guidance, unchanged across three architectures: re-round when shipping a bare quant (the zero-byte serving lever); alternate (§2.4) when a corrector will be attached; never stack re-rounding under a corrector.

3.4 The Lloyd null: further surrogate descent buys nothing (E26)

Would a servable Lloyd-style refinement beat single-pass re-rounding? The natural upgrade alternates the code sweep with a per-row generalized-least-squares refit of the format’s *storable* affine scales (fp16 d/dmin; the 4-bit sub-scales stay frozen so the artifact remains byte-compatible), both halves descending the same damped whitened metric. Run to a pre-registered 8-iteration cap at 0.8B, the alternation cuts the whitened surrogate a further 2.6% below single-pass re-rounding — and the end-to-end effect is statistically zero: paired $\Delta\text{NLL} +0.0009 \pm 0.0022$ ($t = +0.40$), $\Delta\text{KLD} +0.0015 \pm 0.0034$ ($t = +0.43$), $\Delta\text{top-1} -0.55 \pm 0.42$ points ($t = -1.32$), on a harness that resolves the re-round-versus-bare comparison at $t = -27..-47$ [E26]. No codes fixed point was reached within budget (92/96 tensors in a ~0.1% limit cycle — the measured amendment to §2.4’s Claim 3). Together with §3.2 this is the cleanest instance

of the campaign’s recurring lesson: single-pass re-rounding *is* the deployment recipe, because below some floor whitened-proxy descent stops predicting KL divergence. §4.6 collects every instance.

4. Results II: the measured walls

This section is an anatomy of negative results. Each wall is named, measured, and given a mechanism; several were the campaign’s most statistically secure findings [review].

4.1 Wall zero: neither W nor its residual is low-rank (E01/E03)

The spectrum census over all 197 tensors of Qwen3-0.6B: capturing 95% of Frobenius energy needs 50-80% of full rank on every projection type; the compression ratio at that rank is ≈ 1.0 (best: attn_q at 1.09x; V and FFN below 1.0 — factorization *costs* memory there). The stable-rank column (50-200 of 1024) explains why the “LLMs are low-rank” intuition persists: the top of the spectrum is concentrated, but 95% of energy lives deep in a long fat tail [E1]. The quantization residual is worse: r90 $\approx 62\%$ of full rank on average — *less* spectral structure than W itself, behaving as near-isotropic round-off noise, exactly as the Marchenko-Pastur analysis in LQER ([2]) predicts, and as RILQ ([6]) independently diagnosed for the 2-bit regime: the error is inherently high-rank and layer-local correction saturates — and an unweighted rank-128 F16 correction of a Q2_K base recovers 26-44% of error energy while costing 5.26 bpw, already more than Q4_K [E3]. Everything that follows rides on activation-weighted geometry; the unweighted framing is dead at both ends.

4.2 Capture versus width: from a two-point observation to a fitted form (E33)

Diagonal-whitened rank-64 capture of the Q2_K residual: 0.37 at $d = 1024$ (0.6B), 0.07 at $d = 5120$ (27B) — a fivefold fall for a fivefold width growth, at fixed rank. Three published mechanisms would each predict a fall of this kind: a whitened outlier head of near-constant cardinality (massive-activation/super-weight channels number ~ 6 -100 regardless of width; [25], [26], [27]) diluted by bulk energy growing with d ; CALDERA’s $(1-k/2n)^2$ bound and its own tables showing fixed-rank benefit collapsing $7B \rightarrow 70B$ ([5]); and the intrinsic-LoRA lower bound forcing rank to scale near-linearly with width ([11]).

The two points were the observation; the regression (E33) is the measurement. Recomputing whitened spectra with exact totals for $1,146 \text{ tensors} \times 10 \text{ ranks} \times 3 \text{ whiteners}$ on four models — Qwen3-0.6B and Qwen3.5-0.8B at $d = 1024$, heretic-27B and Qwen3.8-27B at $d = 5120$, all from BF16/Q8-provenance pairs with the extractor’s own whitening (the pipeline reproduces the 0.37 and 0.07 above and matches 1,620 logged captures per tensor to ≤ 0.01) — the linear rule capture = $c \cdot (r/d)$ is the *worst* of six candidate forms under every whitener ($\Delta AIC \approx +12,000$ diagonal, +24,000 full-Gram; held-out 27B prediction off 1.5–3.3 $\times$). Returns to rank are sublinear and width enters more strongly than rank: the best pooled form is capture $\approx a \cdot r^{b_r} \cdot d^{b_d}$ with $b_r = 0.61$ [0.60, 0.61], $b_d = -0.67$ [−0.70, −0.64] (diagonal) and $b_r = 0.44$ [0.44, 0.45], $b_d = -0.59$ [−0.61, −0.57] (full-Gram); r and d do not enter only through r/d ($b_r + b_d = -0.06$ [−0.09, −0.04] and -0.15 [−0.16, −0.13]). At fixed rank and tensor kind, diagonal-whitened capture falls as $d^{-0.8 \dots -1.1}$ across the d_{model} -input kinds (ffn_gate -1.13 [−1.17, −1.09]) and full-Gram capture as $d^{-0.5 \dots -0.8}$ — whitening by the full Gram halves the width penalty. The “constant head + isotropic bulk” form fits $h = 0$: no width-invariant outlier-head fraction is visible in these spectra, so the constant-cardinality-head mechanism above is *not* supported by this regression and needs a direct test. The prefactor is not a constant, and not even a per-whitener constant — but the variable is the *base recipe*, not the model: at diagonal r64, $c = \text{capture} \cdot d/r$ is 5.9 (0.6B), 5.2 (0.8B) and 5.9 (heretic-27B — the two-point value reproduces), and on Unsloth’s shipped dynamic quant of Qwen3.8-27B it reads 10.0 (full-Gram 9.5 / 9.3 / 22.7 / 11.5). Re-quantizing Qwen3.8 from BF16 with exactly heretic’s recipe (one Q2_K step, our whitening imatrix, same 11.0 GB byte class) collapses that difference entirely: heretic and Qwen3.8 are indistinguishable at every kind, rank and whitener (diag r64 0.074 vs 0.073, fc r64 0.284 vs 0.290; 0 of 40 per-kind contrasts exclude zero; identical rank exponents), so two qwen35 models of the same width sit on one curve and the shipped dynamic quant sits on another — dynamic bit allocation plus its own calibration inflates diagonal-visible residual structure by $\sim 70\%$ at r64 and halves full-Gram-visible structure. With the fixed-recipe lane as primary, the pooled exponents tighten (diag $b_r = 0.68$ [0.67, 0.68], $b_d = -0.76$ [−0.79, −0.73]; full-Gram $b_r = 0.37$ [0.36, 0.37], $b_d = -0.52$ [−0.54, −0.50]) and holding out either 27B model now predicts it to $0.9\text{--}1.1 \times$ [E33].

Scope: at fixed kind the across-model width axis is one step ($1024 \rightarrow 5120$), the two 27B models share d, the CIs are tensor-resampling CIs (1,000 cluster bootstraps) and not model-resampling CIs, and the prefactor is a property of (whitener, base recipe, width) — a third width (an 8B at $d = 4096$, or a $\sim 70B$) is required before any exponent is quoted as a law [rollup, review].

What is safely established is the direction and its design consequence: at production widths, deployable ranks are a small fraction of dimension, and correction economics degrade faster than byte economics improve. The published counterpoint sharpens the mechanism: SVDQuant ([28]) makes rank-32 work by *engineering* concentration — smoothing manufactures the outlier spike its low-rank branch then eats — which is exactly the structure a generic quantization residual lacks; without a manufactured spike, tiny rank buys r/d and no more.

4.3 The correction-versus-ladder audit at three dense scales

The decisive comparison is not against round-to-nearest but against llama.cpp’s own imatrix-weighted ladder, with the interpolation control included.

27B: the rung is beaten, the interpolation line is not (E27). The 27B comparison was rebuilt under clean provenance: every base one quantization step from the BF16 reference, a fresh 48-chunk BF16 imatrix and Gram capture, BF16 truth logits, and paired per-chunk statistics throughout [E27]. The geometry-aware flagship is a MIXED base (FFN tensors at Q3_K, where the rank-to-width ratio dooms correction; Q2_K elsewhere) plus a full-covariance rank-128 Q8 correction on the non-FFN tensors (capture 0.42). The interpolation control is a pure-quantization mixed-type base built once at the flagship’s byte point (Q3_K_S with attn_k/v/output promoted to Q4_K and a partial attn_qkv promotion; -0.016% bytes), its bytes predicted in advance from our own measured per-kind sensitivities — the same research program minus the adapter. Results on 100 chunks:

config	GB	PPL	KLD	top-1 %
MIXED bare	12.16	7.235 ± 0.114	0.0967 ± 0.0024	87.2
IQ3_XS	12.26	7.303 ± 0.114	0.0767 ± 0.0021	88.8
Q3_K_S	12.37	7.378 ± 0.118	0.0924 ± 0.0024	87.5
MIXED + correction (flagship)	12.50	7.204 ± 0.114	0.0876 ± 0.0024	87.9
interpolation control	12.50	7.194 ± 0.114	0.0814 ± 0.0021	88.3
Q3_K_M	13.59	7.128 ± 0.113	0.0609 ± 0.0016	89.9

Paired per-chunk verdicts, $n = 100$ [E27]:

pair (A vs B)	$\Delta KLD (A - B)$	t	A better on	$\Delta NLL t$	$\Delta top-1 t$
flagship vs Q3_K_S	-0.0048 ± 0.0017	-2.85	71/100	-5.75	+2.04
flagship vs control	$+0.0063 \pm 0.0024$	+2.62	38/100	+0.41	-2.14
control vs Q3_K_S	-0.0110 ± 0.0015	-7.23	91/100	-7.35	+5.55
flagship vs MIXED bare	-0.0090 ± 0.0011	-8.52	92/100	-1.87	+5.20

The corrected configuration beats the adjacent uniform rung on all three paired metrics, and the same-byte control beats the corrected configuration on KL divergence and top-1. The standing 27B ordering at equal bytes is: mixed-type allocation, then low-rank correction, then the uniform rung. The correction’s increment over its own base is real and large in t (-0.0090 KLD, 92/100 chunks); base bits simply spend the same 336 MB better.

The path to these numbers matters for how the earlier ones should be read. At 20 chunks the flagship-versus-rung comparison was a paired tie ($t = 0.6$) and the control led at $t = 1.9$, under this paper’s own 2σ bar. At 40 chunks the tie was exact (KLD 0.0831 versus 0.0832, $t = -0.04$) and the control’s lead reached $t = +2.26$. The 100-chunk run was pre-registered as final with an expectation of certified parity ($|t| < 2$); the data exceeded that expectation in the correction’s favor, and we report it as measured. Before the rebuild, the legacy tables had shown a 1σ unpaired “margin” for the flagship whose sign flipped under clean provenance [review].

Four further facts. First, MIXED bare loses to Q3_K_S (paired KLD $t = +3.6$ at 20 chunks): the base mix alone does not reach the rung, and it is the correction increment that carries the configuration past it. Second, IQ3_XS has the lowest KL divergence in the byte class from below (0.0767 at 12.26 GB), adapter-free. Third, every rebuilt rung scores 0.002-0.003 KLD better than its legacy requantized sibling; the rebuild changed the imatrix along with

the provenance, and the one rung that was already single-step (Q4_K_M) improved by the same amount, so the improvement is a rebuild tax we cannot attribute to provenance alone. Fourth, the task-level check agrees with the instrument where it can see anything. On the full HellaSwag validation set (10,042 items, identical per configuration) the flagship, Q3_K_S and Q3_K_M score 82.25%, 82.39% and 82.85%; no gap reaches the pre-registered 1.4-point threshold. A post-hoc paired McNemar test on per-item outcomes finds flagship versus Q3_K_S a tie ($z = -0.81$) at a KLD difference of about 0.005, and Q3_K_M ahead of both ($z = +3.48$ and $+2.82$) at a difference of about 0.03, in the same order as KL divergence. So on this task the resolution floor of the task evaluation lies between those two KLD differences, and the contested 12.5 GB comparisons sit below it [journal].

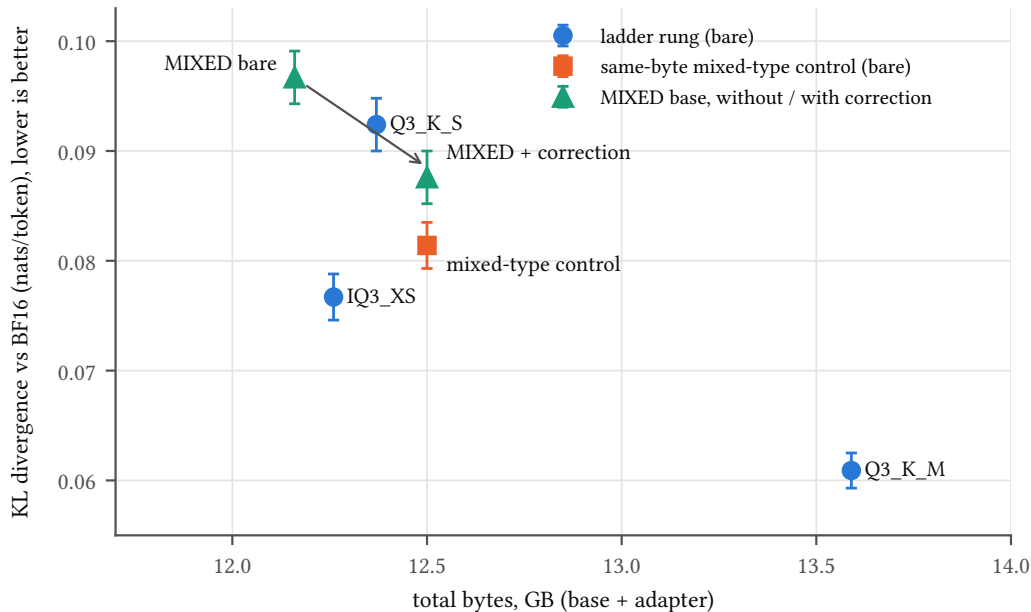


Figure 1: Qwen3.6-27B at the 12.5 GB byte point, 100 chunks, every artifact one quantization step from BF16; error bars are standard errors. The correction (arrow) carries the MIXED base past the adjacent uniform rung Q3_K_S; a pure-quantization mixed-type control at the same bytes is lower still, and the I-quant rung IQ3_XS is the lowest in the class.

One caveat carried from the record: “controls beat corrected configurations at every scale” is an inference at 0.6-0.8B, where no interpolation control was built.

0.6B and 0.8B: wins-from-below. At 0.6B the composed crown (alternation x2 + full covariance + Q8 factors, 382 MB) reaches PPL 25.27 / KLD 0.209 versus the rung below (Q3_K_M, 347 MB, 26.06/0.238 — the KLD margin ~5 sigma on-distribution, surviving Bonferroni over the configs tried) [rollup, review]. At 0.8B, with BF16-truth scoring, fc-r128q8 ties Q3_K_M and one alternation round beats it at +9% bytes (KLD 0.147 vs 0.167, ~3 sigma estimated; PPL only 1.6 sigma) [E22]. But at both scales the comparison rung was chosen from below: the ladder’s own interpolation at the crown’s byte point (≈ 0.118 at 382 MB; ≈ 0.080 at 523 MB) beats the crown on both metrics, and Q4_K_M at +4% bytes is 3-4x better on KLD. No mixed-quant interpolation control was built at small scale; the control we did build at 27B beats the corrected configuration, and the one built for the MoE beats it decisively (§4.5). The honest statement, applied uniformly: the method transfers across architectures and creates points non-dominated by the *discrete* ladder inside the one ~50 MB between-rungs window; the Q4 tier rules above it, everywhere, and the ladder’s interpolation line is unbeaten. E22’s real finding is the transfer itself — the machinery ran unchanged on a second architecture in ~35 minutes on our patched toolchain [E22].

NVFP4 as a carrier, ground-up at 0.8B (E16 rung 1 / E34). The first whitened low-rank correction on a frozen NVFP4 LLM base is real — 40/40 chunks, KLD 0.207 $\rightarrow$ 0.104, top-1 +6.2 points — and loses: the 603 MB composite trails every bare rung at or below its bytes (Q4_K_S at 519 MB, 0.047; Q5_K_S at 578 MB, 0.015), the fourth base family under the equal-byte law of §4.3b. Under llama.cpp’s quantizer, which has no NVFP4-specific importance path, NVFP4 is the *worst* 4-bit carrier on this hybrid-GDN architecture (bare 0.207 against Q4_0’s 0.088 at equal bytes), and its marginally higher residual capture (0.736 versus Q3_K_M’s 0.721) does not buy a larger increment

— Q3_K_M recovers 70% of its damage to NVFP4’s 50%. Capture is a statement about what the factors can see, not about where the composition lands [E34].

Two asterisks on all of the above. Every corrected configuration pays a stock-llama.cpp decode tax of 20-40% that the bare rungs do not (§5); “competes with the rung” and “stock llama.cpp” are never true simultaneously today. And the tables in this subsection are on-distribution: calibration and evaluation share wikitext-2 (train/test split only), and §4.7 shows the method is distribution-sensitive. The held-out evidence for the correction increment at 27B is the FineWeb-Edu control of §4.3c; the ladder comparisons themselves have not been repeated on a held-out corpus.

4.3b The equal-byte equivalence class: mechanism doesn’t matter, allocation does (E28/E29)

A pre-registered afternoon of paired experiments at 27B asked whether *where* the low-rank bytes act changes what they buy. Three mechanisms were built at matched bytes against the same flagship: the post-hoc whitened correction (§2.2, the flagship itself), a pre-quant whitened carve-out in the SRR/preserve-then-quantize order ([29]) — carve $C = B \cdot A$ off the BF16 weights, quantize the deflated remainder with the identical mixed recipe, serve C as the adapter — and shared-basis variants sharing one A factor across input-sharing tensor groups (the GlowQ shape, [30]). Their *surrogate* captures differ wildly: 40% of the residual, 76% of the whitened weight matrix at $k=128$ (E29’s finding that E01’s raw-SVD full-rank null does not transfer to the whitened metric at 27B), +12.2% union capture for the shared basis at parity rank. Their measured quality is indistinguishable: SRR-carve vs flagship KLD $t = +0.99$, shared-A parity vs flagship $t = -0.91$, all $|t| < 2$ on KLD/NLL/top-1 at 100 paired rows each [journal]. The capture surrogate overstating a lever is one more surrogate-versus-outcome divergence (§4.6); SchurQuant’s published observation — “[l]ayer-wise reconstruction is a loose surrogate for the final language-model loss” ([10] §1; their abstract adds that at higher precision “tighter reconstruction does not consistently improve end-model metrics”) — independently corroborates the pattern. Meanwhile the mixed-TYPE control beats every class member ($t \geq +2.3$) and the uniform rung sits below (flagship over Q3_K_S, $t = -2.85$). The synthesis, stated as this paper’s claim: **at 27B, at equal bytes, {post-hoc correction, pre-quant carve, shared-basis} form an equivalence class — the byte budget and its type allocation, not the low-rank mechanism, set quality.** Scope stated per the nearest published duel (AWSRC, [31], whose repair-codec winner flips between PPL and KL at 3B): *among low-rank mechanisms, at 27B, on K-quant grids, at equal bytes.* The codec-class membership question is measured, not open: E31’s pre-registered capture duel on the cached whitened spectra (18 tensors $\times$ 3 seeds) finds AWSRC-class seeded bases capture 0.035 of dominant whitened energy at byte parity versus 0.983 for learned factors — exactly $1.00\times$ the random-subspace expectation, i.e. statistically indistinguishable from random bases in whitened geometry at 27B — so seeded repair sits strictly *below* the equivalence class on head capture and ties only on the isotropic tail (E03); their 3B win is attributed, as a registered testable hypothesis, to low-bit coefficient packing plus an unwhitened low-rank baseline [E31]. The escaping axis is precisely the one RCO ([32]) optimizes — per-tensor type allocation under exact byte budgets, calibration-KL-objective, at 8B — and their paper contains no mechanism-class comparison: our measurement and their optimizer are complementary halves of one statement — allocation is the only lever that moves at equal bytes, and it can be optimized exactly. One deployable residue survives the null: the shared-A basis at r128 ties the flagship’s quality at -16.6% adapter bytes (280.8 vs 336.5 MB, KLD $t = -1.48$), a real VRAM saving under the loader-alias patch class [journal].

4.3c Trained versus one-shot frozen-grid refinement: the GSQ-RCO head-to-head at 27B (E32 / T1.17, third family)

The frozen-grid family of §2/§7 now has a published, artifact-level competitor on consumer formats: IST-DASLab’s GSQ-RCO ([8], [32]) ships refined GGUF checkpoints of Qwen3.8-27B — Gumbel-Softmax re-selection of the discrete codes on shipped K/IQ grids, with per-tensor type allocation optimized under an exact byte budget: per their card, every tensor is GSQ-quantized at every candidate GGUF type into a database, RCO selects one type per tensor under the budget, and the selections are stitched into a standard GGUF. The calibration corpus of the released files is not documented, and the public GSQ repository contains no GGUF path at all (its quantizers train the group scales as free floats), so the comparison is scoped to the *released artifacts*, not to a mechanism ablation. Per the family placement in theory-L6 (Claim 2), this is *search-and-granularity*, not surrogate-versus-outcome: at layer granularity GSQ optimizes the same quadratic our closed-form pass solves, by stochastic relaxation instead of one

exact whitened solve. The head-to-head was pre-registered (experiments/32-t117-gsq-head-to-head/README.md) on the same shipped Unsloth Dynamic baselines their model card compares against, graded on 40 chunks of wikitext-2 test against BF16 truth logits regenerated for this family, with a fresh 48-chunk BF16 imatrix+Gram capture (the E27 recipe unchanged on a hybrid Gated-DeltaNet architecture — the third architecture family the machinery has run on) [E32].

On wikitext KL divergence the one-shot pass wins at both rungs. Our r128-Q8 whitened correction on the bare Unsloth IQ2_S base reaches KLD 0.1191 ± 0.0027 (top-1 85.8 %) at $8.37 + 0.90$ GB against GSQ-RCO IQ2_XS’s 0.2028 ± 0.0044 (81.4 %) at 8.42 GB — paired $\Delta\text{KLD} -0.0837 \pm 0.0073$, $t = -11.5$, 40/40 chunks, top-1 +4.4 pt ($t = +10.9$). At the IQ3 rung — their “task-lossless” operating point — ours reaches 0.0353 ± 0.0011 against 0.0525 ± 0.0012 : $\Delta\text{KLD} -0.0173 \pm 0.0018$, $t = -9.5$, 38/40 chunks, top-1 +1.6 pt ($t = +4.7$). One row up, on this instrument, GSQ-RCO lands *below its own untrained baseline* — the bare Unsloth quant at the same byte class beats the trained refinement at both rungs (IQ2: $+0.0715$ KLD, $t = +11.7$, 0/40 chunks in GSQ’s favor; IQ3: $+0.0152$, $t = +8.6$, 1/40). The correction increment over its own bare base is real at all three rungs measured (IQ2_S -0.0122 , $t = -6.5$, 39/40; Q2_K_XL -0.0055 , $t = -7.0$, 34/40; IQ3_S -0.0021 , $t = -5.9$, 36/40), with the top-1 increment real at the two 2-bit rungs ($+0.7$ pt, $t \approx +3.8$) and tied at IQ3_S ($t = +0.5$). The cost asymmetry is large: about 13 CPU-minutes per model, no GPU, on the machine that serves the result, against a multi-GPU training run.

The equal-byte law survives its third family. Byte-fair against the ladder rather than the competitor, the result is §4.3’s again: the bare Unsloth Q2_K_XL at 9.83 GB beats our corrected IQ2_S-plus-adapter at 9.27 GB by 0.0337 KLD ($t = +11.6$, 0/40). Refinement of either kind — trained or closed-form — repairs a rung from within; interpolation between rungs at equal bytes still wins, and GSQ-RCO, which carries an allocation optimizer (RCO) for exactly that axis, sits below the unrefined ladder on this instrument as well. This is the statement of §4.3b extended to a third family.

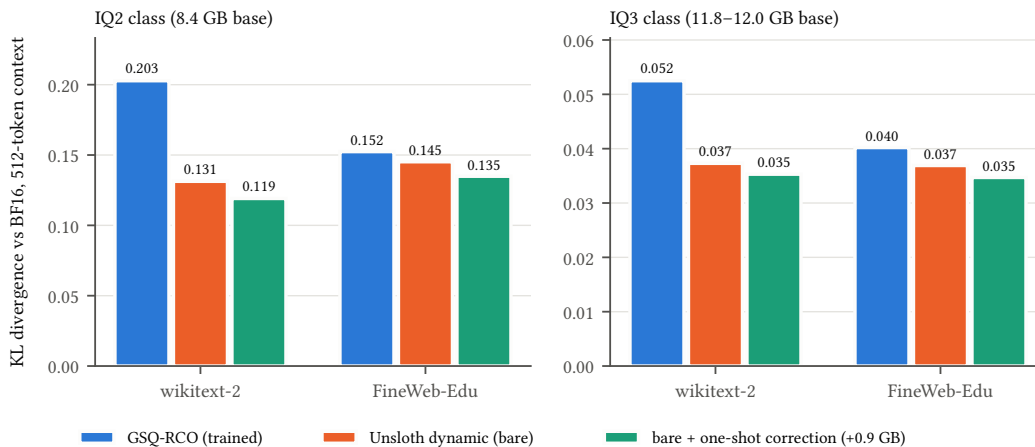


Figure 2: Released GSQ-RCO checkpoints of Qwen3.8-27B against their untrained Unsloth baselines and against the baseline plus a one-shot whitened correction, on two corpora (40 chunks each). The corrected arm carries a 0.9 GB adapter and is not byte-matched to the other two.

The held-out control: the correction transfers, and the competitor’s deficit mostly does not. The wikitext table leaves a calibration-mismatch reading open: the released files’ calibration corpus is undocumented, so their artifact may be tuned away from our instrument rather than worse in general. We pre-registered the control — the same eight rows graded on 40 chunks of FineWeb-Edu, the one non-wikitext corpus their card reports, against a BF16 reference on that text — with the prediction that under the mismatch reading the GSQ-RCO deficit against the bare baseline would shrink toward zero, and under the alternative would stay at $t \geq +4$ [E32].

pair, FineWeb-Edu, n = 40	Δ KLD (A - B)	t	A better on
GSQ-RCO IQ2_XS vs bare UD-IQ2_S	+0.0073 $\pm$ 0.0044	+1.66	13/40
GSQ-RCO IQ3_S vs bare UD-IQ3_S	+0.0033 $\pm$ 0.0016	+2.07	13/40
ours (IQ2_S + correction) vs GSQ-RCO IQ2_XS	-0.0175 $\pm$ 0.0045	-3.94	35/40
ours (IQ3_S + correction) vs GSQ-RCO IQ3_S	-0.0055 $\pm$ 0.0016	-3.39	31/40
ours vs own base, IQ2_S	-0.0103 $\pm$ 0.0011	-9.40	38/40
ours vs own base, Q2_K_XL	-0.0052 $\pm$ 0.0006	-9.20	38/40
ours vs own base, IQ3_S	-0.0022 $\pm$ 0.0003	-8.65	36/40
ours (IQ2_S + correction) vs bare UD-Q2_K_XL	+0.0471 $\pm$ 0.0044	+10.72	1/40

The mismatch reading is supported for the competitor: its deficit against the untrained baseline falls roughly tenfold at the IQ2 rung (from +0.0715 at $t = +11.7$ to +0.0073 at $t = +1.66$) and fivefold at IQ3, to the edge of resolution. Our own increments do not shrink: a correction whitened by wikitext Grams improves its base on FineWeb-Edu by nearly the same absolute amount as on wikitext, at higher t , at all three rungs. This is the paper’s held-out confirmation of the correction increment at 27B. The one-shot arm still leads the trained arm at both rungs, by a smaller margin, and the ladder still beats the corrected configuration at near-equal bytes.

Context length reverses the perplexity ordering. The card’s wikitext perplexities order GSQ-RCO ahead of the Unsloth baseline at both rungs; our 512-token rows order them the other way. Re-measured at a 2048-token context, the card’s ordering reproduces: GSQ-RCO IQ2_XS 6.834 ± 0.043 against UD-IQ2_S 7.119 ± 0.047 , and GSQ-RCO IQ3_S 6.306 ± 0.039 against UD-IQ3_S 6.417 ± 0.041 [E32]. The discrepancy is therefore neither a build nor a tokenizer artifact. The trained refinement is better than its baseline on long-context perplexity and worse on short-context divergence from the reference. Every KL-divergence number in this paper is measured at 512 tokens.

Capability reverses the divergence ordering. The card’s claims are task-level, so the decisive comparison for their arm is a task suite. We ran the 112-unit fast subset of an agentic coding suite (multi-language tasks solved by an autonomous agent loop with a compiler and tests; single-slot serving, stock llama.cpp b10865, reasoning budget capped at 20,480 tokens) on the two IQ3-class artifacts [E32]:

artifact	passed (of 112)	completion tokens
GSQ-RCO IQ3_S	88	2.40 M
Unsloth UD-IQ3_S (bare)	76	2.75 M
reference band: an uncensored finetune of the same model at Q5_K_M, four runs	82-91	—

Paired by unit: 16 units pass only under GSQ-RCO and 4 only under the baseline, a net of +12 (exact McNemar $p = 0.012$); the direction holds in both the Zig subset (+7, $p = 0.065$) and the rest (+5, $p = 0.18$). The trained artifact lands inside the Q5-class band, which is what the card’s “task-lossless at IQ3_S” predicts, and the untrained baseline lands below it. This is one run per arm; the single-slot run-to-run floor of this suite was not measured, and the first attempt at these rows, under six-slot serving, was retracted in full after both IQ3-class artifacts crashed the server (a CUDA launch timeout that Q5-class files did not trigger in 14 hours). The IQ2-rung capability pair was started and not completed, and our own corrected arm was not run on the suite.

What the head-to-head establishes. On divergence from the BF16 reference at 512 tokens, the one-shot whitened correction beats the released trained artifacts at both rungs and on both corpora, and its increment over its own base holds on a corpus it was not calibrated on. On that same instrument the trained artifact scores below its untrained baseline; on long-context perplexity and on agentic coding it scores above. The instrument and the outcome disagree about the competitor, and we have no task measurement of our own arm. We therefore claim the fidelity result as measured and make no claim that the corrected artifact is the more capable one. The comparison also adds an entry to the ledger of §4.6, this time against the instrument this paper relies on: short-context KL divergence resolved real differences within one refinement family throughout this study, and it did not rank artifacts across two.

4.4 The recovery curve and the IQ1 asymptote

Held to a single estimator (full covariance), the fraction of the base’s KLD damage recovered by correction rises monotonically as the base approaches its cliff: 20% at Q2_K, 28% at IQ2_XS, 41% at IQ1_S [rollup, review]. (The

steeper 7.5%→18%→41% curve in early notes mixed diagonal and full-covariance points and is retired. A residual confound the metric fix does not cure, disclosed per the ablation audit: even the fc-consistent curve varies adapter budget — 866/866/799 MB — and rank mix across its three points, so no point holds the correction recipe fixed while varying only the base; that rebuild has not been run.) The curve’s qualitative content matches ResComp’s inverse-bit-width claim ([33]) and EoRA’s biggest-at-2-3-bit gains: correction pays most exactly where pure quantization collapses.

The asymptote test takes this to its limit — the founding pure-factorization hypothesis in its last servable form: the thinnest carrier llama.cpp has (IQ1_S, 1.56 bpw, 7.44 GB) with factors doing maximum representational work. Scaling the corrector to r512 non-FFN / r256 FFN (2.52 GB of Q8 factors, BF16-pure residuals) lifts recovery to 52% — and still loses to bare Q2_K at near-equal total bytes by ~3x KLD (10.50 PPL/0.431 at 9.96 GB versus 7.891/0.153 at 10.86 GB). The exchange rate is brutal and worsening: the adapter’s second and third gigabytes bought ~0.10 KLD in *total* (0.527→0.431 over +1.72 GB, i.e. ~0.056 per GB) where a gigabyte of carrier bits buys ~0.28 — a ~5× contrast [E15]. *Factors cannot substitute for carrier bits*. The constructive corollary is the campaign’s design rule (§4.8); the destructive one is the epitaph of pure factorization at calibration-only budgets, agreeing with the published price list for real factorization ratios (50B+ retraining tokens: Sheared-LLaMA [34], Minitron [35]).

The corrected IQ1_S point itself (8.24 GB file, KLD 0.891→0.527, top-1 +8.5 points, 16.5 sigma) is the one place the ladder offers no measured rival rung — with the review’s caveats attached: the equal-byte rivals IQ1_M and IQ2_XXS were never built, and 8.24 GB is file bytes, projecting to ~10.3-10.5 GB VRAM on the measured adapter-overhead pattern [review].

4.5 The MoE wall: the pooled expert residual is nearly isotropic — no shared subspace to amortize (E23)

On Qwen3.6-35B-A3B (256 experts/stack, 8 routed; expert stacks are 91% of parameters), both correction variants are real and both lose the ladder decisively — with the interpolation control this time built in from the start [E23]. Reference disclosure, per §2.5’s targeted-reference rule: no BF16 of this model exists on our hardware, so every row (and the KLD scoring) is one quantization step from the strongest vendor artifact (UD-Q4_K_M, 22.1 GB) — uniform provenance within the table, with every verdict margin $\geq 2.7\sigma$ paired and the losing margins at 6.9-8.2 σ , comfortably outside the reference-contamination scale measured at 27B. Per-expert whitened correction (r32, Q8, pooled-Gram whitening; served on *stock* llama.cpp — build_lora_mm_id handles 3D adapter pairs natively, a fact we established by measurement) improves the pinned-Q2_K base at 4.6 paired sigma (18/20 chunks). But at its own byte point (14.5 GB) it loses to the *trivial* control — promote `ffn_down_exps` to Q4_K — by $+0.0257 \pm 0.0038$ KLD at 6.9 sigma, 0/20 chunk wins; Q3_K_S at +0.7 GB is ~2x better again. Per byte spent, quant-type promotion buys ~4x more KLD reduction than rank-32 factors at this scale.

The sharper negative is structural. The shared-basis hypothesis — one output basis B per stack amortized across 256 experts, motivated by all experts reading/writing the same activation spaces — fails *at the capture level*: the shared basis captures 0.086-0.090 of whitened residual energy on up/gate stacks (gate 0.090, up 0.086) against a random-orthonormal floor of 0.0625, and 0.022 versus 0.0156 on down stacks. Stated precisely: 256 subspaces cannot be literally mutually orthogonal in a 512-dim output space — what the shared-capture-at-the-floor measurement shows is that the *pooled* whitened residual spectrum is nearly isotropic, with no fat shared subspace to exploit at $r = 32$ [E23]. Mechanistically the whole MoE wall is the *r/d* observation in miniature multiplied by expert count: each expert is a small 512x2048 matrix whose Q2→Q4 residual is nearly unstructured (per-expert capture 0.19-0.36 versus 0.7+ at dense small scale — measured under the pooled whitener, a disclosed approximation of each expert’s conditional input covariance, so some of the “unstructured” reading may be whitener mismatch, mostly on down stacks), and thin-factor overhead is paid 256 times. Routes that could change the verdict — correcting the *dense* tensors of the MoE, or sub-2-bit expert bases where the ladder has no cheap promotion rung — are follow-ups, not rescues.

4.6 Estimator gates metric: the allocation and two-sided evidence

A recurring lesson of the campaign is that metric *quality* dominates rank, and estimator quality dominates metric sophistication — and, at the floor, that surrogate improvement stops predicting outcome at all. Positive direction: diagonal → full covariance at 0.6B was worth more than doubling rank (fc r64: PPL 27.58 / KLD 0.318 versus

diag r64 31.29/0.420 and diag r128-Q8 28.46 — ~ 10 sigma and 2.6 sigma respectively), and the same-pipeline full-versus-diagonal ablation at 27B resolved at ~ 2 sigma (0.1460 $\rightarrow$ 0.1282 at r64) [E10]. Negative direction, instance by instance: (i) greedy energy water-filling for rank allocation is actively harmful (34.17 versus uniform’s 31.29 at equal bytes — it zeroes `ffn_down`, whose 25,600-dim inputs hold the massive-activation outliers the diagonal proxy cannot see); measured per-kind KL sensitivities fix the ordering (`attn_k` dominates at 0.0128 Δ KLD/MB, additivity verified: Σ isolated 0.352 $\approx$ joint 0.346), though the allocation-versus-uniform end-to-end margin at 27B is itself unresolved (0.5 sigma) and the 27B “measured” weights were transferred 0.6B priors — relabeled accordingly [journal, review]. (ii) Gradient-side-only sensitivity anti-correlates with measured sensitivity (Spearman -0.29) exactly as input-only does; only the two-sided product flips positive (+0.39), and the leftover-correlation structure after diagonal whitening is *inverted* between the two sides per kind (grad-side headroom in attention K/V, input-side in `attn_output/ffn_down`) — either single side is a trap, and two-sided whitening is the only lane that collects both [E20]. Measured two-sided fc r64 at 0.6B: 27.27 versus 27.58 one-sided at equal bytes. (iii) The E24 Kronecker result (§3.2): the *exactly-solved* richer metric with a noisily-estimated factor is worse than the poorer metric estimated well, at 15 paired sigma. (iv) The E26 Lloyd null (§3.4), the cleanest instance of all: a 2.6% further descent of the *same* whitened surrogate — no estimator change, no metric change — produced exactly zero end-to-end effect ($|t| \leq 1.3$ on all metrics) on a harness that resolves real effects at $t = 27-47$. The full ledger, with the instances reported in other sections:

#	what improved on the surrogate	what the outcome did	where
1	energy captured by greedy rank allocation	PPL worse than uniform (34.17 vs 31.29)	§4.6 (i)
2	exact descent of a two-sided Kronecker metric, measured T	+0.048 nats vs input-only ($t = +15.5$); worse than bare with a clean T	§3.2
3	further 2.6% descent of the same whitened objective (Lloyd)	no change ($ t \leq 1.3$)	§3.4
4	importance moved into column scales (equalization)	PPL 33.09 $\rightarrow$ 41.73 ($t = +28$ to $+60$)	§4.6b
5	whitened capture at equal bytes (shared basis, +12%)	KLD tie ($t = -0.91$)	§4.3b
6	residual capture of the base (NVFP4 0.736 vs Q3_K_M 0.721)	smaller increment (50% vs 70% recovery)	§4.3
7	512-token KL divergence across refinement families	ordering reversed on 2048-token PPL and on agentic coding	§4.3c

Rows 1-6 are surrogates for KL divergence that failed to predict KL divergence. Row 7 is KL divergence itself failing to predict a task outcome across families, after predicting it within one (§4.3, HellaSwag). The lesson in one sentence: *an approximation of the metric is not a monotone approximation of the outcome, and below some floor whitened-proxy descent stops predicting KL divergence at all.*

4.6b The equalization null: importance belongs in the fit weights, not the columns (E26)

The AWQ/SmoothQuant lineage moves importance into the weights: scale columns by an importance factor so the quantizer spends its dynamic range where activations are large. On an *imatrix-weighted* quantizer, we measured the full-strength version of this idea as the strongest-t negative in the paper. Method integrity first: on this architecture only *diagonal* gauge transformations fold exactly (every candidate rotation site sits behind an RMSNorm with learnable gamma), and restricting the scales to powers of two makes the fold a bit-exact gauge transformation — verified end-to-end on GPU, equalized BF16 versus stored BF16: KLD 0.000000, top-1 100.000% — a reusable methods nugget (pow2 folds are exact gauge transformations; the imatrix transforms exactly with the gauge). Requantizing the equalized model with the identical recipe at identical bytes is then strongly anti-helpful: PPL 33.09 $\rightarrow$ 41.73, paired $t = +28..+60$, equalization better on 0/40 KLD chunks; the quantizer’s own imatrix-weighted objective degrades 6-68% per tensor (eq/bare error ratio 1.06-1.68 on folded tensors, exactly 1.000 on the untouched control) [E26]. The defensible mechanism statement: moving importance from the fit weights into the column scales is strongly harmful for Q2_K’s shared-sub-scale format — the fold injects up to $32\times$ dynamic range into 16-element sub-blocks that the shared 4-bit sub-scales cannot span, and flattens the informative weighting of the affine fit. Whether an “importance double-counting” component exists independently of the format interaction is *not* discriminated by this run (in exact arithmetic the weighted objective is gauge-invariant, and the equalized run counts importance once — in the columns); the discriminating controls (equalization without imatrix; an alpha sweep, where the AWQ lineage predicts an interior optimum) have not been run. As measured, the result is scoped

to the pre-registered full-strength point ($\alpha = 1$) — precisely where the equalization literature itself expects harm — on one format.

4.7 Task conditioning: the surviving form of the factorization dream

Fixing the target distribution collapses functional rank. Conditioning the correction’s covariance on a code corpus buys +10.3% on-distribution perplexity at identical bytes (code-conditioned 2.746 versus wiki-conditioned 3.061 on code, ~ 8 sigma) at a steep off-distribution cost (34.90 versus 27.58 on wiki — a 27% swing), with asymmetric generalization: broad calibration transfers to narrow far better than the reverse [E17]. This is simultaneously a product lane (per-workload adapters via llama.cpp’s per-request LoRA API) and the sharpest internal caveat: the measured distribution sensitivity of the method (10-27%) exceeds every contested ladder margin in §4.3, which is why §3.1 and §4.3c report held-out corpora and why the ladder comparisons of §4.3 are labeled on-distribution.

4.8 The co-design ordering rule

Assembled from §3.3 and the alternation arc, measured at two scales and paired-confirmed at a third: *joint shaping beats greedy sequential composition*. Final 0.6B co-design ranking: alternation x2 (25.27) > ProjQ-deflated one-shot re-round (25.94) > greedy re-round-then-correct (26.80), with the alternation lineage disclosed (LoftQ-descended; the novelty is the black-box llama-quantize grid, the whitened metric, and the termination certificate of §2.4) [E13, review]. Together with the geometry rule — correct where rank is a large fraction of width; spend base bits where it is not, validated constructively by MIXED at 27B and destructively by E15 and E23 — this is the paper’s prescription to a practitioner: the order of operations is a measured result, not a convention.

5. The systems layer: pricing and removing the serving tax

5.1 The tax, stated first

Every quality result in §4 that involves an adapter carries a cost the bare quant rungs do not. On stock llama.cpp the LoRA path adds two unfused GEMV launches per projection, and batch-1 decode pays for them: -33% to -39% at 0.6B (667 $\rightarrow$ 444 and 407 tok/s) and about -20% at 27B. The published record shows the same pathology elsewhere: adapter operations are rank-independent and launch-bound at batch 1 (Punica, [36]; AdaFuse, [37]; FlashSVD v1.5’s 1,174 launches per token, [38]; CALDERA’s Table 6, [5]). Under the protocol of §2.5 this tax is priced beside every byte-fair claim: “competes with the rung on stock llama.cpp” is not currently a true sentence. The kernels below remove most of the tax on local, unmerged patches, are digit-identical in perplexity, and are kill-switched at every stage. All speed verdicts are same-session A/B pairs. Between-session GPU boost drift reached 9% on identical configurations and invalidated one headline cell from our first tables (86.3 tok/s, 82.3 on re-measurement); for the same reason the “-45% $\rightarrow$ -20%” of §5.2 is stated against that session’s own bare baseline and differs from the 0.6B pairs quoted here [review].

5.2 Three phases of kernel fusion

Phase 1 — window fusion. The 5-node subgraph llama.cpp emits per adapted projection ($\{W \cdot x, A \cdot x, B \cdot t, \text{scale}, \text{add}\}$) is pattern-matched in the CUDA backend and collapsed to two kernels: $B \cdot t + \text{scale} + \text{add}$ ride the quantized-GEMV (mmvq) epilogue. Result: digit-identical PPL, +10% at r16, exact parity at r64 — and a falsified premise. Removing 588 of 784 extra kernels/token moved r64 by ~ 0 : under Blackwell CUDA-graph replay, tiny launches were nearly free. Under GPU contention, however, the same fusion was worth +42% (single uncontrolled run; direction real, magnitude unreproduced) [E14].

Phase 2 — fold A into the activation quantize; find the real wall. The design template is SVDQuant’s ([28]): fold $A^T x$ into the activation-quantize stage and $B \cdot t$ into the GEMM epilogue, the fusion that took their low-rank branch from +50% latency to 5-10% — re-derived here for the GEMV/batch-1 regime their diffusion-serving engine does not cover. Folding $t = A \cdot x$ into the pre-existing quantize_q8_1 launch (dedicated trailing thread blocks; no atomics, no new graph nodes) plus a redesigned epilogue took r64 decode from 476 to 685 tok/s (+44%), collapsing the adapter tax from -45% to -20%, rank-flat and bit-identical. The diagnosis that paid: the constant cost was never launch overhead but *tail-load serialization* — the epilogue loaded B and t after the main dot loop, adding a full memory-latency round to every mmvq’s critical path (and four serial rounds on the small-K variant). The

generalizable idiom: **prefetch static operands before the PDL grid-dependency sync** — weights can legally stream into registers while `cudaGridDependencySynchronize` waits for the producer grid, converting exposed latency into overlap. A long-standing rank-64 speed anomaly dissolved with the same fix [E14].

Phase 2B — quantized factors; fusion is not free once the rival is dp4a. Accepting Q8_0 factors in both fused paths gave +62% at 0.6B (394 → 638) but initially *regressed* 27B by 6.6%: unfused Q8 B-t already rides an int8 dp4a kernel upstream, while the fused epilogue’s per-row float tax grows with model size and launch savings do not. Two empirical size gates (rows x rank; fold-K) restored a no-regression +0.4% at 27B — an honest crossover measurement rather than a win [E14].

Phase 3 — quantized-t sidecar and the occupancy law. The planned fix (a q8 sidecar of t enabling an int8 dp4a epilogue) was built and is digit-clean — and turned out to be the small half. The real 27B tax was *register pressure*: the fused instantiation carried ~19 extra registers (61 → 80), dropping resident blocks from 4 to 3 for every fused W kernel across ~2.8M output rows/token. Splitting the LoRA template off the GLU fusion machinery and moving the entire adapter dot before the main loop — its result parked in shared memory across the existing barrier, no adapter register live in the main loop — restored lean occupancy (62 registers). Final same-session pairs: 0.6B 391 → 702 tok/s (+79%); 27B 84.7 → 86.9 (+2.6%, a single pair). Re-measured on a quiet host with interleaved blocks (N = 10, median and interquartile range), the 27B flagship decodes at 88.1 [88.0-88.3] fused against 85.8 [85.7-85.8] unfused: +2.7% with disjoint ranges, against a measured ceiling of +4.9% for any epilogue [E27]. The second transferable idiom: **epilogues of memory-bound GEMV kernels must be register-neutral in the main loop; shared memory is the only safe place to carry epilogue state.** A corollary with deployment value: launch savings scale inversely with clocks, so fused margins grow under contention and thermal limits — the regime real multi-tenant serving lives in [E14].

5.3 The allocator fix: +19.6% for everyone

llama.cpp’s GGML_CUDA_GRAPH_OPT stream concurrency was being correctly *rejected* on adapter rigs: ggml-alloc keeps one global free-list, so a buffer freed inside one fork branch is handed to a tensor in a concurrent branch — a genuine cross-stream write-after-read hazard (an earlier interleave attempt corrupted generation silently). The fix is allocator-level branch-window pinning: a tensor flag marks fork-join regions; frees inside a region are deferred to the join; an in-place guard prevents the last consumer of a fork tensor from stealing a buffer other streams still read. Cost: zero measured buffer growth. Verification: the full backend test suite (12,996 cases) passes; PPL digit-identical and 200-token greedy generation *string-identical* on/off across all three rigs — the check the earlier corruption escaped. Result: 0.6B base decode 794 → 950 tok/s (+19.6%, reproducing the Phase-2 prize with events validated rather than lucky); +7-12% with the adapter under residual host load; 27B within noise (streams cover only the hybrid’s 1-in-4 full-attention layers, and 5120-10240-row GEMVs already fill the GPU — concurrency pays on launch-bound small kernels). A bonus root-cause: the 27B rig had been forming zero stream regions because pair-seeding rejected Q8_0 A factors — absence of concurrency logs is a distinct failure mode from invalid events [E25]. This fix is unrelated to compression and benefits any CUDA llama.cpp user with graph optimization; with the GRAPH_OPT interleave-restore bug found and fixed in Phase 2 (a silent 552-tok/s regression mode), it forms the upstream-shaped half of the patch set [wins].

5.4 Serving-tax accounting after fusion

Where the ledger stands: at 0.6B the adapter tax on our patched build is ~20% (down from 45%), and the free +19.6% base-model lever stacks on top; at 27B the flagship’s fused decode (88.1 tok/s) sits within a bounded +4.9% ceiling of unfused, i.e. the tax at production scale is already small once factors are Q8 — the remaining gap to bare-rung serving is the adapter’s extra VRAM (~0.7 GB class) and the residual per-token latency chain, which the megakernel literature prices at 1.3-2 μs/node (Mirage MPK, [39]; Hazy megakernels) and which allocator-level concurrency, not further fusion, is the identified next lever. Every number in this section comes from the per-phase reports at a pinned build (llama.cpp 0ef6e55ed), with kill switches and same-session pairing. The patch set has since been rebased onto llama.cpp b10865 and is vendored with the experiment tree; no speed number here was re-measured after the rebase, and all of them are from one GPU [E25].

6. Limitations and open work

6.1 Limitations

The campaign was reviewed adversarially twice — statistics, claims, code, and then the paper draft itself [review] — and the limitations that survive are the paper’s boundary conditions.

Distribution. Calibration and most evaluation share wikitext-2 (train/test split only). The corrected configurations carry far more calibration-fitted capacity than the rungs they compete with, and §4.7 measures 10-27% corpus sensitivity, larger than every contested ladder margin. What we have off-distribution is: the re-rounder at 0.8B on a code corpus (worse than bare when calibrated on wikitext alone; better when code is in the calibration mix) and on an unseen prose corpus (better in both variants) (§3.1); and the correction increment at 27B on FineWeb-Edu, where it holds at three rungs (§4.3c). The ladder comparisons of §4.3 — flagship versus rung versus control — have not been repeated on a held-out corpus, and neither has the 27B re-rounder.

The instrument. Every KL-divergence number is measured at a 512-token context. §4.3c shows two artifacts of one model whose ordering under that instrument reverses on 2048-token perplexity and on an agentic coding suite. Within a single refinement family the instrument agreed with the one task evaluation that could resolve anything (HellaSwag, §4.3); across families it did not. Task-level evidence in this paper is thin: one multiple-choice benchmark at one byte point, and one coding suite, one run per arm, on the competitor’s rung only. None of our own corrected or re-rounded artifacts was run on the coding suite.

Controls. The mixed-quant interpolation control exists at 27B and for the MoE, and was never built at 0.6B or 0.8B. This is the reason §4.3 claims wins-from-below at small scale rather than wins.

Width. The capture regression of §4.2 has one step on the width axis ($1024 \rightarrow 5120$) and its confidence intervals resample tensors, not models.

References and precision. Rows labeled legacy are referenced to a Q6_K or Q8_0 file rather than BF16 (measured contamination at 27B: 0.0032 KLD). The 27B endgame is single-step from BF16 and paired at 100 chunks; the remaining 27B lever comparisons (measured allocation versus uniform rank, alternation rounds) are inside unpaired error bars, the 27B VRAM columns were not measured, “full covariance” at 27B means block-diagonal for `ffn_down` (§2.2), and the 27B rank allocation used sensitivity priors transferred from 0.6B.

Evidence grade on alternation. The ordering claim of §4.8 (never stack re-rounding under a corrector) is paired-resolved ($t = +17.6$). The claim about alternation *rounds* is weaker: the 0.6B arc is unpaired, the 0.8B continuation has no recorded error bars, and the 27B increment from alternation is about 1σ .

Speed. All speed numbers come from one GPU. Session-to-session drift reached 9%; kernel gates are tuned on two model sizes; nothing was re-measured after the patch set was rebased (§5.4).

Code hazards, disclosed. Cache directories originally carried no provenance fingerprint (silently wrong on resume; fixed for the re-rounder, open for the spectra caches and one extractor). Column selection under Gram whitening ranks the wrong columns (latent; the reported column runs predate full-covariance whitening). A naive MoE Gram accumulation would have been wrong and was caught before any reported capture [review].

How the work was done. Experiments, analysis and drafting were carried out by AI research agents under the author’s direction, on one workstation. That is why the review rounds and the pre-registration rule exist, and it did not prevent errors: this record contains a set of paired t-statistics that were inflated by dividing per-chunk differences by a per-token standard error (caught and recomputed with the canonical tool before any conclusion changed), two capability rows that were retracted when 85% of their failures turned out to be a crashed server, and a first “mixed-calibration” run that never reached its second corpus. Each is kept in the journal with its correction.

6.2 Status of the ablation program

An audit of every lever-at-scale cell behind this paper’s claims counted 51 cells: 17 clean, 15 confounded, 19 missing [ablation]. The nulls are the cleanest cells (water-filling, shared basis, equalization, the Kronecker metric, MoE versus control — isolated, mostly paired, 2.7 to 60σ). The confounds concentrate in the positive multi-point curves: the recovery curve of §4.4 varies adapter budget and rank mix across its points, and the 27B allocation row varied three factors at once.

The audit priced a matrix of seventeen runs to close the gating cells. Eleven have been run and are reported above: the 100-chunk BF16 reference; the paired 27B re-round (§3.1); the single-step 27B rebuild with its control, at 20, 40 and 100 chunks (§4.3); the first task evaluation (§4.3); the capture-versus-width regression (§4.2); the third-corpus held-out test and the full-metric mixed-calibration run (§3.1); the clean-Gram Kronecker run (§3.2); and the GSQ-RCO head-to-head on two corpora with one capability pair (§4.3c). Six have not been run, and the corresponding claims are scoped accordingly in the text:

- the calibration-sensitivity grid (mix ratio $\times$ method), which would turn §3.1’s rescue demonstration into an ablation;
- interpolation controls at the 0.6B and 0.8B byte points;
- a held-out pass over the corrected configurations and their ladder rivals at all scales;
- the recovery curve with the correction recipe held fixed;
- the equalization mechanism ablations (no imatrix; an alpha sweep);
- a bound on calibration-sampling variance alone.

Also open: the IQ2-rung capability pair and a capability run of our own arm (§4.3c); a single-slot run-to-run floor for the coding suite; a 27B re-round pair with single-step provenance; and a third model width.

6.3 Reproducibility and artifacts

Method descriptions, scripts, raw logs, per-chunk outputs and result tables for every experiment are in the project repository under `research/lowrank/experiments/`, one directory per experiment, and Appendix A maps each tag used in this paper to its directory. Large binaries — reference logits, Gram captures, quantized models and adapters, about 160 GB — are not in the repository. They are rebuildable from the recorded recipes, and some early ones were deleted to reclaim disk, so a number resting on a deleted artifact is reproducible only by rebuilding it. Re-runability was measured twice: one reproduction gate rebuilt the 0.8B base 335/335 tensors byte-identical from the recorded recipe and re-derived the re-rounded artifact byte-identical, and another reproduced the 0.8B published numbers exactly before any new comparison was run.

Three components are not stock `llama.cpp`: the Gram-capture extension to `llama-imatrix` (about 140 lines, environment-gated), the `llama-gradmatrix` tool, and the fused-kernel and allocator patch set (reported numbers at build `0ef6e55ed`; vendored rebased onto `b10865`). The re-rounder and all extractors are standalone Python against `gguf-py`. The served artifacts need none of the patches: a re-rounded file is a standard GGUF, and a corrected model is a standard GGUF plus a standard LoRA-form adapter.

6.4 Open lanes

Codec coverage for the free lever. The re-rounder speaks only Q2_K, Q3_K and Q4_K codecs would reach the promoted tensors that mute the 27B result, I-quant codecs would allow a like-for-like head-to-head with GSQ, and an NVFP4 codec (nearest-of-8 E2M1 on frozen FP8 scales) targets a format whose residual is covariance-structured ([40]). The intended tool is `gguf-refine model.gguf`, shipping with a mixed-calibration default and a held-out validation gate as requirements: §3.1’s measurement makes an ungated version of this tool a quality hazard [wins].

Estimator repair for two-sided metrics. The ranked follow-ups from §3.2: damping or shrinkage of the output-side factor toward the identity (does the harm vanish smoothly?), multi-pass gradient captures, and a delta-net adjoint to remove the gradient cut. More coverage of the same single-pass estimator is ruled out by the clean-Gram run.

Geometry. Of the differential-geometry candidates we surveyed [manifolds], the survivors are ProjQ-style metric surgery (already positive: 25.94 PPL one-shot at 0.6B) with its Bregman-damped variant ([21]), Riemannian refinement of the closed-form factors on fixed-rank manifolds, and the information-geometry reading of the whitening ladder. The Kronecker global metric is measured anti-helpful at current estimator quality, and closed-form Fisher allocation with one-sided proxies is a confirmed trap.

Product lanes. Per-workload conditioned adapters (+10.3% on-task at equal bytes, selectable per request through the existing LoRA API); a sparse-A fused kernel (top correction directions live on 2-5% of input channels); the shared-A adapter (−16.6% adapter bytes at tied quality, §4.3b); correction of the dense tensors of an MoE and

sub-2-bit expert carriers; and the upstream-shaped bundle — the re-rounder with its Gram-capture extension, the allocator fix, and the fused kernels — each useful on its own.

6.5 Closing statement

The ledger of this study: one lever that is free at fixed calibration breadth, paired-resolved at 0.8B and at 27B; a correction paradigm that is real, composable and convergence-certified, that beats the adjacent ladder rung at 27B and loses to a same-byte mixed-type control, and whose three low-rank forms are interchangeable at equal bytes; a kernel and allocator patch set that helps users who will never load an adapter; a head-to-head in which our instrument favored our arm and a task suite favored the competitor over its own baseline; and a protocol that would have prevented our first week of overclaims. The quantization ladder that ships in llama.cpp is close to its local optimum, and type allocation is the axis along which it can still be improved at equal bytes. The codes on its frozen grids are not at their optimum. That asymmetry — the walls around the corrections, the freedom inside the grids — is the finding.

7. Related work

The correction lineage. Adding a low-rank branch to a quantized base begins with ZeroQuant-V2’s LoRC ([1]), where an unweighted rank-8 SVD acts as stabilizer rather than fixer; LQER ([2]) states our premise in published form — the raw quantization residual is Marchenko–Pastur-shaped, so plain-SVD capture is $\approx r/d$ and *weighted* decomposition is where the value lives — and QERA ([3]) supplies our reference mathematics: its Theorem 1 is the full-covariance whitened closed form we use, and its Theorem 2’s diagonal special case is *literally llama.cpp’s imatrix statistic*, which is why our pipeline’s diagonal-to-full-covariance ladder is exactly the QERA-approx→QERA-exact step measured at production scale. EoRA ([4]) is the closest industry neighbor (eigenspace projection $\equiv$ QERA-exact; $\sim 3\times$ full-vs-diagonal at 3-bit), and CALDERA ([5]) contributes both the joint alternation-with-best-iterate template we adopt and — in its own tables — the fixed-rank benefit collapsing with scale (r64 buys 0.86 PPL at 7B; r128 buys 0.05 at 70B). What none of this lineage reports, and we add: same-pipeline diagonal-versus-full ablations on *deployed* K/I-quant formats at 0.6B–35B, comparisons against the format’s own quantization ladder rather than round-to-nearest, and mixed-quant *interpolation controls* at matched bytes — the control that, at 27B and MoE scale, our own corrections lose to (§4.3).

Rank must scale with width. The capture decay we measure and fit in §4.2 (a fitted form over four models and one width step, not a law) has three published legs: CALDERA’s Theorem 4.1 bound, the Bid-Up/OLrC lower bound ([11]) forcing fixed-rank share to decay $\sim 2r/d$, and the empirical twins — LRC ([41]), where $r/d = 10\%$ halves the W4A4 gap and 30% closes it; ResQ ([42]), which hard-codes $r = d/8$ in production; FLRQ ([43]), whose allocated average rank 39 beats uniform r256 at a fifth of the bytes; and TwinQuant ([44]), measuring LLaMA3-8B spectra flat beyond rank 256. RILQ ([6]) publishes the plateau’s diagnosis — 2-bit error is inherently high-rank and layer-local SVD saturates — which our whitened 27B numbers confirm on shipped grids.

Frozen-grid discrete refinement. Re-choosing a quantized model’s codes on its shipped grids is a small, recent family whose members are — provably — descent procedures on one layerwise quadratic (the L6 note derives each placement). GPTQ ([7]) is the ancestor: its error-feedback sweep is Babai’s nearest-plane algorithm, and on frozen grids Bid-Up proves per-sweep non-increase. ReQuant ([9]) performs exact cyclic coordinate descent on the same objective (its $-\Delta q \cdot g + \Delta q^2 \cdot \tilde{H}_{jj}$ score is the exact per-coordinate change; verified from its Eqs. 5–10), with a GPTAQ-style quantized-prefix asymmetry that reduces to our symmetric case at $\Delta X = 0$ — generic PTQ, not GGUF. SchurQuant ([10]) unfreezes the scale block: suffix-eliminating Schur conditioning (the chunk-block generalization of the Babai sweep’s column conditioning) alternated with closed-form continuous scale/zero refits, worth +11.88pp at 2-bit with the objective held at GPTQ’s — per-row-group format, not the K-quant superblock hierarchy. GSQ ([8]) is the only other member operating on *shipped GGUF artifacts*: Gumbel-Softmax relaxation trained ~ 20 GPU-epochs against block-staged reconstruction, projecting back into format (Qwen3-8B Q2_K task average 50.03→56.28). Our re-rounder is the one-shot, whitened, CPU-cheap point in this family — in-format on the shipped superblock hierarchy, KLD-graded. §4.3c reports the artifact-level head-to-head against the released GSQ-RCO checkpoints, with our correction rather than our re-rounder as the one-shot arm (the re-rounder’s codec does not yet cover the I-quant formats those checkpoints use); the result is a fidelity win for the one-shot arm

and a capability win for the trained artifact over its own baseline. SchurQuant’s own finding that “[l]ayer-wise reconstruction is a loose surrogate for the final language-model loss” (§1) joins our surrogate-versus-outcome series (§4.3b) as independent corroboration.

Metric sophistication and its ceiling. The whitening ladder ascends from the imatrix diagonal through the full input Gram (ours, GPTQ, QERA-exact) toward two-sided and end-loss metrics: GuidedQuant ([45]) reaches the best pure-2-bit numbers in the lineage on block-diagonal end-loss Fisher *without any correction branch* — the ceiling our metric-ladder results point toward — and OBD-LLM ([17]) applies the two-sided Kronecker-whitened objective to post-GPTQ residual correction, of which our objective is the $L_g = I$ special case. We defend one-sidedness rather than apologize for it (§2.2): the tool path starts from a shipped GGUF where no gradients exist; the cheap diagonal end of output weighting is provably a no-op for frozen-grid rounding (E24’s negative theorem); and their measured two-sided gain is on weight decomposition at 8B, not residual correction on K-quant grids. Their 10%-mean-diagonal dampening recipe (verified verbatim) and KronQ’s HG-cancellation result ([46]) are the adoptable details; ARHQ’s residual-covariance whitening ([47]) is the metric variant our cross-layer proposal inherits for free.

The equivalence class, its floor, and its escape. Where the low-rank bytes act turns out not to matter: §4.3b’s pre-registered triple measurement — post-hoc correction, SRR-style pre-quant carve ([29]), GlowQ-style shared bases ([30]) — is byte-fair-indistinguishable at 27B despite wildly different surrogate captures. The class has a measured floor: AWSRC’s seeded-Hadamard codec ([31]), the nearest published byte-fair repair duel, captures exactly the random-subspace expectation of dominant whitened energy in our geometry (E31: 0.035 vs 0.983 at byte parity) — strictly below every low-rank member, tied only on the isotropic tail. And it has one measured escape: per-tensor type allocation, which is precisely the axis RCO ([32]) optimizes with exact-budget dynamic programming at 8B — a paper containing zero mechanism-class comparison. The two results are complementary halves of one statement: *allocation is the only lever that moves at equal bytes, and it can be optimized exactly*. FLRQ and LQ-LoRA’s ILP ([48]) allocate rank and bits respectively under the same doctrine.

Mixture-of-experts. MiLo ([13]) is our exact architecture on MoE — HQQ base plus truncated-SVD compensators with INT3 factors — and states its compensator is unfused; TileQ ([49]) tiles within experts, consistent with our finding that pooled expert residuals are near-isotropic (§4.5): no shared subspace exists to amortize, and per-expert correction loses to a trivial quant-promotion control at 6.9 paired sigma. The router-guided compensator family ([50]) spends FLOPs we don’t have; we are byte-bound.

Serving is part of the claim. The field’s artifacts conspicuously lack fused decode paths: EoRA ships unfused in HF/vLLM (its GPTQModel kernel is the lone exception), CALDERA’s Table 6 records the decode loss, MiLo declares its compensator unfused, and SVDQuant’s Nunchaku ([28]) — the fusion template our kernels follow, with its measured naive +50% latency collapsing to 5–10% fused — ships low-rank-over-NVFP4 for diffusion only. Our fused-GEMV trilogy and the priced serving-tax ledger (§5) exist because a correction that decodes 20–40% slower on stock llama.cpp is not the same artifact as its bare base, and the lineage’s quality tables silently assume it is.

The upstream conversation. llama.cpp discussion #8831 proposed an LQER-style correction in 2024 and was met with the prediction (ik_llama.cpp #15) that mature quants would not benefit; this paper is a systematic empirical answer to that thread — the skeptic was mostly right about correction versus the ladder, and wrong about the grid itself, whose frozen codes carry a free 13–34% of KL divergence (§3). Two of our campaign’s independent findings (the MTP-imatrix blind spot; the KV-backward seam) were already reported upstream (#23476/#23575; #21037) and are treated here as confirmations, per our stated novelty protocol (§2.5).

Measurement stance. We grade artifacts by mean KL divergence against reference logits with paired per-chunk statistics — KLD because it tracks answer flips at Spearman 0.981 ([22]) and is the deployment community’s working currency (llama.cpp #4110) — with perplexity secondary under a stated silent-zone caveat ([23]). Our own §4.3c is a counterexample to leaning on it across refinement families, and we report it as one. The same-harness survey’s warning that cross-paper numbers are incomparable without shared calibration and evaluation is one this paper takes as a design constraint rather than a disclaimer: every comparison inside is same-pipeline, same-tokens, same-reference.

Appendix A. Artifact index

Bracketed tags in the text name the place in the project repository where the measurement, its raw logs and its scripts live.

tag	path
E1	research/lowrank/experiments/01-svd-spectrum/
E3	research/lowrank/experiments/03-residual-rank/
E10	research/lowrank/experiments/10-full-covariance/
E13	research/lowrank/experiments/13-rerounder/
E14	research/lowrank/experiments/14-fused-kernel/
E15	research/lowrank/experiments/15-iql-carrier/
E17	research/lowrank/experiments/17-task-conditioned/
E20	research/lowrank/experiments/20-gradgram/
E22	research/lowrank/experiments/22-qwen35-08b/
E23	research/lowrank/experiments/23-moe/
E24	research/lowrank/experiments/24-yaqa-lite/
E25	research/lowrank/experiments/25-alloc-concurrency/
E26	research/lowrank/experiments/26-lloyd-gauge/
E27	research/lowrank/experiments/27-bf16-rederivation/
E31	research/lowrank/experiments/31-seeded-basis-duel/
E32	research/lowrank/experiments/32-t117-gsq-head-to-head/
E33	research/lowrank/experiments/33-t110-capture-width/
E34	research/lowrank/experiments/34-e16-nvfp4-08b/
ablation	research/lowrank/paper/ABLATION-PLAN.md
journal	research/lowrank/JOURNAL.md
manifolds	research/lowrank/prior-art/MANIFOLD-CANDIDATES.md
protocol	research/lowrank/PROTOCOL.md
review	research/lowrank/review/
rollup	research/lowrank/RESULTS_ROLLUP.md
theory-L6	research/lowrank/paper/theory-L6-family-subsumption.md
theory-alt	research/lowrank/paper/theory-alternation-convergence.md
wins	research/lowrank/paper/DEPLOYABLE-WINS.md

References

Works are identified by arXiv number; short names are as recorded in the project bibliography (research/lowrank/paper/references.md), which also records what each source was used for.

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- [5] CALDERA (NeurIPS 2024). arXiv:2405.18886.
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- [40] Bridging the Gap for Microscaling FP4 (ISTA; ICLR 2026). arXiv:2509.23202.
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- [42] ResQ (Purdue/d-Matrix; ICML 2025). arXiv:2412.14363.
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